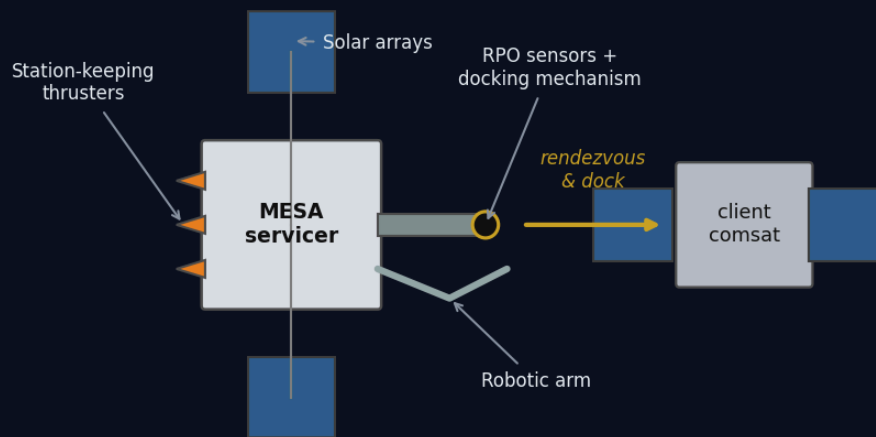


MESA

Mission Extension and Servicing Asset for GEO

MESA: GEO Mission Extension and Servicing Asset



SPCE 5065: Space Environment Interactions, Design Project Final Report

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Executive Summary

MESA, the Mission Extension and Servicing Asset, is a 2,000 kg geostationary servicing tug that docks with client satellites in the GEO belt, takes over their stationkeeping and attitude control, refuels them, and tows them between the operational belt and the graveyard orbit. It is built for a five-year contract inside a \$100M cap, and it is patterned on flight-proven servicing heritage rather than on a paper concept [1].

The case for this design rests on four results, which are derived in this report rather than asserted.

GEO's dominant hazards are spacecraft charging and energetic-particle radiation, not drag or debris. MESA floats to -2.16 kV during substorms and accumulates 25 krad(Si) over five years behind 100 mils of aluminium. Both are handled by conventional, qualified means: a fully bonded conductive exterior, single-point grounding, and radiation hardness assurance category R parts at 100 krad(Si), which holds even if the dose assumption is wrong by a factor of two.

Docking, not free flight, drives the design. Capturing a 3,000 kg client raises the roll inertia by a factor of 5.3, which is what sizes the reaction wheels at 0.20 N·m rather than the 3.21×10^{-5} N·m disturbance torque. More importantly, docking bonds two independently charged vehicles through a capture latch, a failure mode no conventional satellite carries. MESA answers it with a plasma contactor and an operational rule that equalizes potential before mechanical contact.

Every budget closes with margin. Mass closes at 2,000 kg with 150 kg of margin, 11 percent of dry mass. Power closes with 68 percent margin over peak load at end of life. Propellant closes with the hydrazine tank as the binding sub-budget, which still supports about ten client servicing cycles against a baseline manifest of six. Cost closes at exactly the \$100M cap with \$5M held in program reserve.

The analysis is independently verified. All results come from one reproducible Python model, and the orbit, eclipse, array output, and orbital lifetime were rebuilt independently in Systems Tool Kit. Where the two agree, the report says so; where a check is a consistency check rather than an independent one, it is labelled as such.

Table ES-1: MESA at a glance.

Quantity	Value	Where
Orbit	GEO, $a = 42,164$ km, $e = 0$, $i = 0^\circ$	Eq. (1)
Wet mass / dry mass	2,000 / 1,400 kg	Table 4
Mass margin	150 kg (11% of dry)	Table 4
Array output, BOL to EOL	3,387 W to 2,984 W	Eq. (16)
Peak electrical load	1,773 W	Table 5
Bus temperature, sun and eclipse	+36.3 °C and -13.1 °C	Eq. (7)
Floating potential in a substorm	-2.16 kV	Eqs. (8) and (9)
Total ionizing dose, five years	25 krad(Si)	Eq. (10)
Worst-case disturbance torque	3.21×10^{-5} N·m, 93.5% from SRP	Eq. (11)
ADACS mass and power	157 kg, 283 W	Eq. (13)
Five-year delta-v	463 m/s	Eq. (4)
Servicing capacity	10 client cycles, hydrazine-limited	Eq. (5)
Program cost	\$100M, including \$5M reserve	Table 8

Nomenclature

Symbols are grouped after the acronyms. Units are given for every dimensional quantity; a dash marks a dimensionless one.

Acronyms

Acronym	Meaning
ADACS	Attitude Determination and Control System
AO	Atomic oxygen
BOL / EOL	Beginning of life / end of life
CME	Coronal mass ejection
DoD	Depth of discharge
EDAC	Error detection and correction
EP	Electric propulsion
ESD	Electrostatic discharge
EUV	Extreme ultraviolet
GCR	Galactic cosmic ray
GEO	Geostationary Earth orbit (~35,786 km altitude)
ITO	Indium tin oxide (transparent conductive coating)
LEO	Low Earth orbit
MEO	Medium Earth orbit
MESA	Mission Extension and Servicing Asset (this system)
MEV	Mission Extension Vehicle (Northrop Grumman)
MLI	Multi-layer insulation
MMOD	Micrometeoroid and orbital debris
OSR	Optical solar reflector
RHA	Radiation hardness assurance
RPO	Rendezvous and proximity operations
SAA	South Atlantic Anomaly
SEU / SEL / SEGR	Single-event upset / latch-up / gate rupture
SEP / SPE	Solar energetic particle / solar proton event
SRP	Solar radiation pressure
SWPC	(NOAA) Space Weather Prediction Center
TID	Total ionizing dose
TVAC	Thermal-vacuum (testing)

Symbols

Symbol	Meaning	Units
a	Semi-major axis	km
A	Area, subscripted by the surface it refers to	m ²
A_{eff}	Sunlit array area integrated by the STK Solar Panel tool	m ²
A_s	Illuminated area, the SRP and drag reference area	m ²
B	Earth magnetic flux density at the orbit radius	T

Symbol	Meaning	Units
c	Speed of light, 2.998×10^8	m/s
c_{ps}, c_{pa}, c_m	Center of solar pressure, of aerodynamic pressure, of mass	m
C	Lumped thermal capacitance	J/K
C_D	Drag coefficient	-
D	Residual magnetic dipole	A·m ²
e	Orbital eccentricity	-
E_{batt}	Installed battery capacity	W·h
F	Cumulative meteoroid flux above a given particle mass	m ⁻² yr ⁻¹
F_s	Solar constant at 1 AU, 1,361	W/m ²
g_0	Standard gravity, 9.807	m/s ²
G	Gravitational focusing factor	-
G_s	Solar irradiance used by STK, 1,361.128	W/m ²
H	Stored angular momentum	N·m·s
i	Inclination, and solar incidence angle in Table 12	deg
I	Moment of inertia about the slew axis	kg·m ²
I_x, I_y, I_z	Principal moments of inertia	kg·m ²
I_{sp}	Specific impulse	s
k_B	Boltzmann constant, 1.381×10^{-23}	J/K
m	Spacecraft mass	kg
m_0, m_p	Mass at the start of a maneuver, propellant consumed by it	kg
M	Control torque for a slew maneuver	N·m
M_E	Earth magnetic moment, 7.96×10^{15}	T·m ³
M_{RW}	Reaction wheel torque requirement	N·m
p	Probability of at least one impact	-
P	Electrical load or solar array output power	W
P_{orb}	Orbital period	s
q	Surface reflectance factor	-
q_e	Elementary charge, 1.602×10^{-19}	C
Q_{Earth}	Absorbed Earth infrared plus albedo load	W
Q_{int}	Internal dissipation, the electrical load rejected as heat	W
r, R	Orbit radius measured from Earth's center	km, m
R_a	Radius of the shielding atmosphere, $R_E + 100$ km	km
R_E	Earth equatorial radius, 6,378	km
S	Solar intensity, zero in umbra to one in full sun	-
t	Time, or the duration of a maneuver	s
t_b, t_w	Whipple bumper and rear wall thickness	cm
T	Temperature; a subscripted T in Table 12 is a torque	K
T_e	Plasma sheet electron temperature	K
T_{sp}, T_g, T_m, T_a	SRP, gravity gradient, magnetic, and aerodynamic torque	N·m
T_D	Worst-case total disturbance torque	N·m
v	Orbital velocity	m/s

Symbol	Meaning	Units
V	Spacecraft floating potential	V
α	Solar absorptivity	-
Δv	Velocity increment	m/s
ε	Infrared emissivity	-
η	Solar cell conversion efficiency	-
θ	Off-nadir angle, or slew angle	deg
μ	Earth gravitational parameter, 3.986×10^{14}	m ³ /s ²
ρ	Atmospheric (neutral) density	kg/m ³
ρ_p	Meteoroid particle density	g/cm ³
σ	Stefan-Boltzmann constant, 5.670×10^{-8}	W/m ² K ⁴
χ	Earth-shielding factor	-

1. Introduction

1.1 The mission

The GEO belt is full of satellites that are healthy in every respect except one: they have run out of stationkeeping propellant. A communications satellite whose transponders will work for another decade gets retired because it can no longer hold its slot, and the operator writes off an asset worth far more than the fuel it lacks. That is the market MESA is built for.

MESA, the Mission Extension and Servicing Asset, is a 2,000 kg geostationary servicing vehicle that flies a rendezvous with a client satellite, docks with it, and then flies the combined stack: providing attitude control and stationkeeping for a client whose own propulsion is gone, transferring propellant to refuelable clients, and towing satellites between the operational belt and the graveyard orbit roughly 300 km above it. The concept follows Northrop Grumman's Mission Extension Vehicle, which launched MEV-1 in October 2019 and docked Intelsat 901 in February 2020, taking over that client's stationkeeping [1]. MESA broadens that proven single-purpose vehicle into a reusable servicing asset that handles multiple clients in sequence across a five-year contract, inside a \$100M program cap.

1.2 What is in this report

This report is the complete design case for MESA, from the environment it must survive to the hardware that survives it.

Sections 2 and 3 define the system and its orbit: what MESA does, who buys it, how a servicing cycle actually runs, and why the orbit is geostationary. Sections 4 through 6 characterize the environment: the Sun-Earth system and the hazards it creates at GEO and at LEO, the space weather that modulates them, and the vacuum environment that makes ground testing non-negotiable. Section 7 defines the vehicle itself and closes the mass, power, propellant, and cost budgets, which is where a design either works or does not.

Sections 8 through 11 are the subsystem design. Section 8 sizes the thermal control system and gives the equilibrium temperatures in sun and in eclipse. Sections 9 and 10 work the plasma and radiation environments in turn, each covering the risks to this vehicle, the proposed mitigations, and what those mitigations cost the other subsystems. Section 11 estimates the four external disturbance torques and sizes the attitude control hardware from them.

Section 12 presents the supporting simulations: the orbit, the power profile across a full day including eclipse, the thermal transient through that eclipse, and an independent rebuild of the mission in Systems Tool Kit. Section 13 pulls the whole design together into a single risk picture, adding the micrometeoroid and debris assessment and the docking failure modes, and Section 14 states the conclusions.

The through-line, and the reason this design is worth the contract, is that a GEO servicer is not a communications satellite with an arm bolted on. It is charging- and radiation-limited rather than drag-limited, its attitude control is sized by the client it captures rather than by the disturbances it feels, and the single most dangerous moment in its mission lasts about a second.

1.3 Relationship to the earlier milestones

This report is cumulative. Milestone 1 established the environment and chose the orbit, Milestone 2 added the subsystem design, and both are carried forward here in corrected form. Appendix A records the corrections made to Milestone 2 and the work added since.

2. Satellite System Name and Mission Objectives

2.1 System name and rationale

The system is named **MESA (Mission Extension and Servicing Asset)**. The expansion is a deliberate nod to the heritage it follows, mirroring Northrop Grumman's Mission Extension Vehicle while broadening the scope from a single life-extension client to a reusable asset that also refuels, repairs, and relocates [1]. As a word, a mesa is a high, flat-topped highland that commands a view of the plain below: a fitting image for a vehicle that operates from GEO, the highest common regime, holding a stable perch above the LEO and MEO traffic with persistent overwatch of its clients.

2.2 Customer

The primary customer is the U.S. Space Force and Space Systems Command, which operates high-value national-security assets in the GEO belt and has a standing interest in keeping them on station and in repositioning them without expending their own propellant. The secondary customer base is the commercial GEO communications operators, such as Intelsat and SES, whose satellites routinely outlive their fuel.

2.3 Mission objectives

Primary objectives.

1. **Rendezvous and dock** with a cooperative GEO client satellite.
2. **Provide attitude control and stationkeeping** for a client whose own propulsion is depleted or degraded, acting as a bolt-on propulsion and ADACS module and extending the client's operational life.
3. **Transfer propellant** to refuelable clients.
4. **Tow** defunct or low-fuel satellites between the operational GEO belt and the graveyard orbit for refuel or repair, and return them to a working slot.

Secondary objectives.

1. **Relocation service:** reposition healthy clients to new longitude slots on operator request.
2. **Debris mitigation:** perform end-of-life disposal boosts of clients to the graveyard, freeing operational slots.
3. **Inspection:** use the rendezvous sensors to image and diagnose a client's exterior before servicing, which is a saleable product in its own right.

The governing constraints are a servicer life of at least five years, multiple clients serviced in sequence, and a \$100M program cap. These objectives drive the derived requirements that every later section works against: precision RPO sensors (cameras and LIDAR), a docking mechanism and robotic arm, generous stationkeeping propellant, and avionics hardened against the charging and radiation environment.

2.4 Concept of operations

The objectives above only become design drivers once they are placed on a timeline, so **Figure 1** lays out one servicing cycle and the five-year manifest that follows from it.

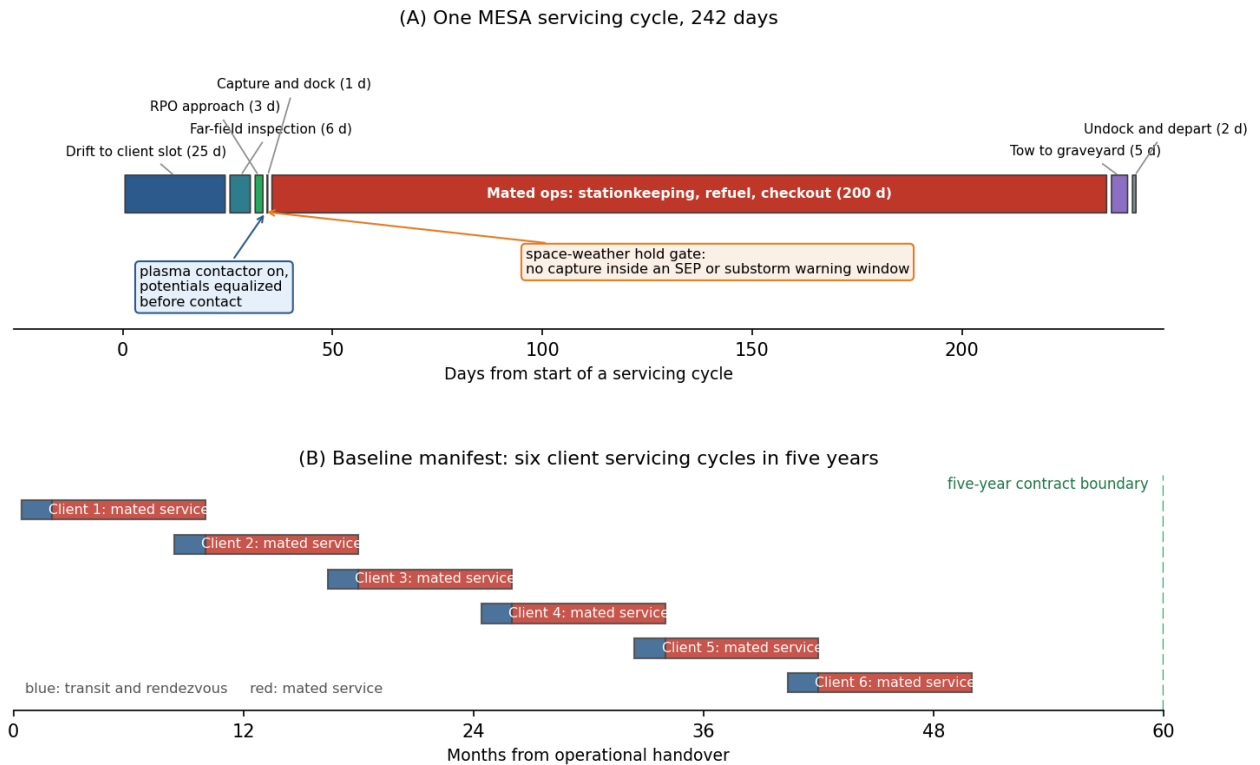


Figure 1: MESA concept of operations. Capture is the short, irreversible step in a long cycle, which is why the design gates it on space weather and equalizes potential first.

A cycle runs about 242 days and is dominated by mated operations. MESA drifts to the client's longitude slot over roughly 25 days, inspects the client from a safe standoff for about six days, closes over three days of proximity operations, and then captures. Capture itself takes about a second of contact inside a day of approach. Mated service runs for the bulk of the cycle, during which MESA holds the combined stack on station and performs the refuel or repair, and the cycle closes with a tow to the graveyard where required, then undocking and departure for the next client.

Two design consequences fall directly out of this timeline and are carried through the rest of the report. First, the vehicle spends about eighty percent of its life mated to a client, so the attitude control system, the thermal interface, and the propellant budget must all be sized for the stack rather than for the free flyer. Second, the single irreversible step is a moment of physical contact between two independently charged vehicles, which is why Section 9 puts a plasma contactor on the vehicle and why capture is gated on the space weather feed described in Section 5.

3. Orbit Selection and Orbital Lifetime

3.1 The selected orbit

MESA flies a geostationary orbit. The orbital elements are given in Eq. (1):

$$a = 42,164 \text{ km (altitude } 35,786 \text{ km)}, \quad e \approx 0 \text{ (circular)}, \quad i = 0^\circ \text{ (equatorial, stationkept)} \quad (1)$$

where a is the semi-major axis measured from Earth's center, e the eccentricity, and i the inclination of the orbit plane to the equator.

The orbit is circular so the tug holds a constant altitude and speed relative to its clients, which is what makes a slow, controlled approach possible, and equatorial at zero inclination, maintained by stationkeeping, so it stays fixed over one longitude. The assigned slot for the analysis in this report is 105° W , one of the two stable longitudes produced by Earth's triaxiality (Section 3.3). Parking on a stable point removes the dominant east-west drift term, so the east-west stationkeeping allowance carried in Section 3.3 is conservative for this slot. The scale and geometry are shown in **Figure 2**.

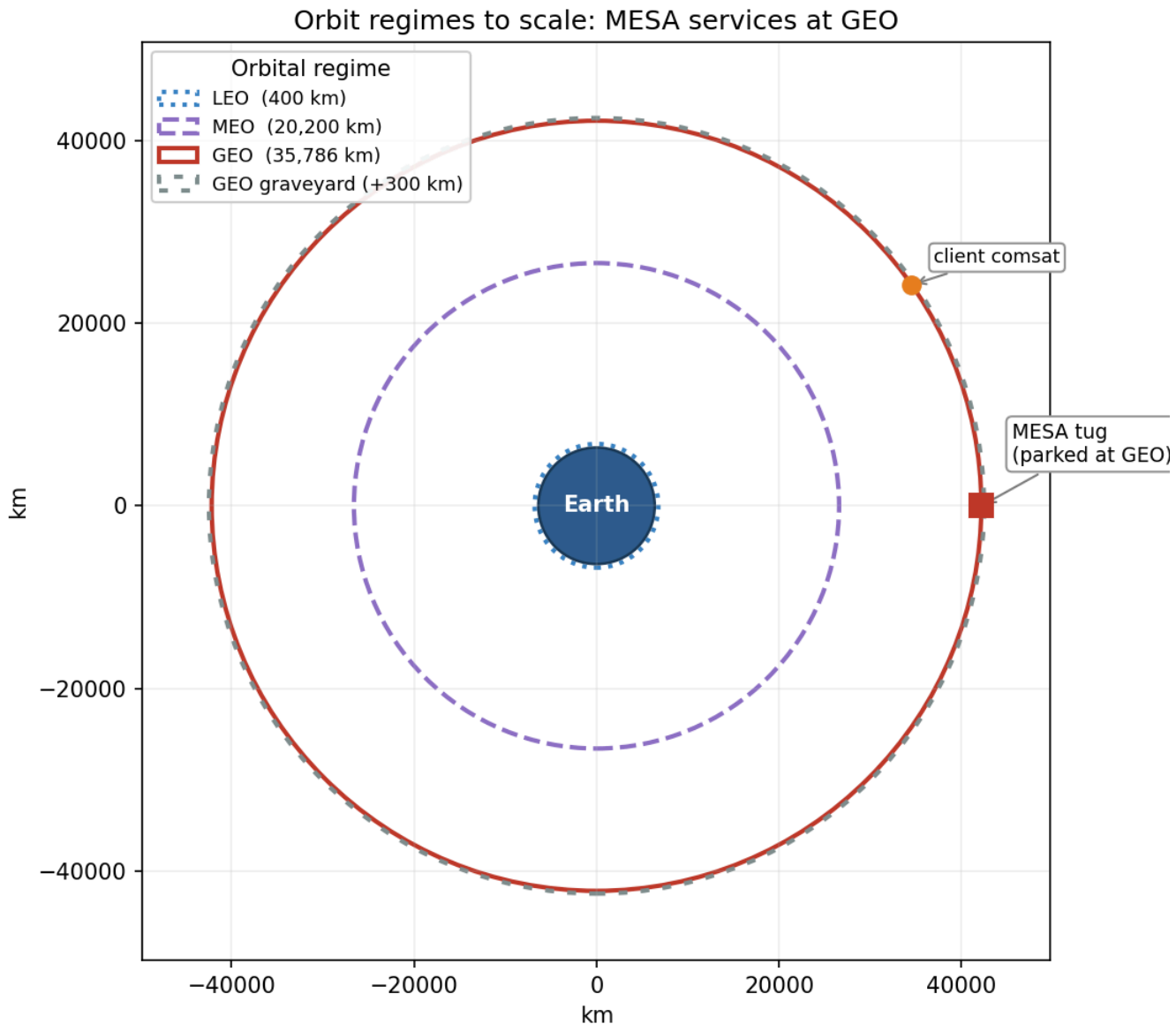


Figure 2: Equatorial view to scale. The tug operates in the GEO belt (35,786 km), moving clients to and from the graveyard.

3.2 Rationale, and why the alternatives fail

The orbit selection is the least negotiable decision in the design, because the clients live at GEO. Defunct and low-fuel communications satellites, national-security assets, and the graveyard orbit about 300 km above the belt are all there, so MESA must operate in the belt to rendezvous, dock, refuel, and tow them [1]. The operational fit reinforces this: a geostationary orbit fixes the longitude, which gives continuous line-of-sight to a fixed ground station and therefore simple telemetry, tracking, and command, alongside access to the dense, high-value GEO customer base.

That decision does carry a real environmental penalty: GEO trades LEO's atmospheric drag, atomic oxygen erosion, and dense debris population for worse spacecraft charging and worse energetic-particle radiation. Sections 4 through 6 characterize both regimes and Section 4.5 works that trade against the evidence.

Neither alternative regime survives scrutiny. LEO fails on the mission itself, because the clients are not there, and it fails on physics too: Section 3.3 shows a tug with MESA's ballistic coefficient decaying out of a 400 km orbit in under a year, and atomic oxygen and debris are both far worse. MEO sits deep in the Van Allen belts, the worst radiation environment of the three, with no client base to justify absorbing it. GEO insertion is expensive, but one tug amortized across many clients fits inside the \$100M and five-year envelope, and the mission is inherently geostationary.

3.3 Orbital lifetime without stationkeeping

The natural next question is what actually ends this mission, and the direct answer is that **at GEO, atmospheric drag is not the life-limiting mechanism**. Using the Homework 2 neutral-density model ($\rho = 1.020 \times 10^7 h^{-7.172}$ kg/m³, with h in km) alongside a standard GEO density of about 10^{-15} kg/m³, the characteristic drag-decay timescale at GEO is on the order of 10^5 to 10^6 years, roughly five orders of magnitude beyond the five-year requirement.

The governing relation is the circular-orbit drag-decay rate, Eq. (2), integrated as $t = \int da/|\dot{a}|$ from the starting altitude down to a 150 km reentry:

$$\dot{a} = -\rho \frac{C_D A}{m} \sqrt{\mu a}$$

(2)

where $\dot{a}$ is the rate of change of semi-major axis (m/s), ρ the neutral density at altitude (kg/m³), C_D the drag coefficient, A the ram cross-sectional area (m²), m the spacecraft mass (kg), μ the Earth gravitational parameter (3.986×10^{14} m³/s²), and a the semi-major axis (m). The grouping $C_D A/m$ is the inverse ballistic coefficient (m²/kg), which is the only vehicle property the result depends on.

To demonstrate the tool on a case where drag actually matters, Eq. (2) was applied to this same vehicle in LEO. Taking the wet mass of 2,000 kg, a ram area of 15 m², and $C_D = 2.2$, so that $C_D A/m = 0.0165$ m²/kg, the decay times are given in **Table 1** and plotted in **Figure 3**.

Table 1: LEO drag-decay case for the MESA ballistic coefficient (Homework 2 model).

Starting altitude	Decay time to 150 km
300 km	28.5 days (0.08 yr)
400 km	298 days (0.82 yr)
500 km	1,836 days (5.03 yr)
600 km	22.2 yr

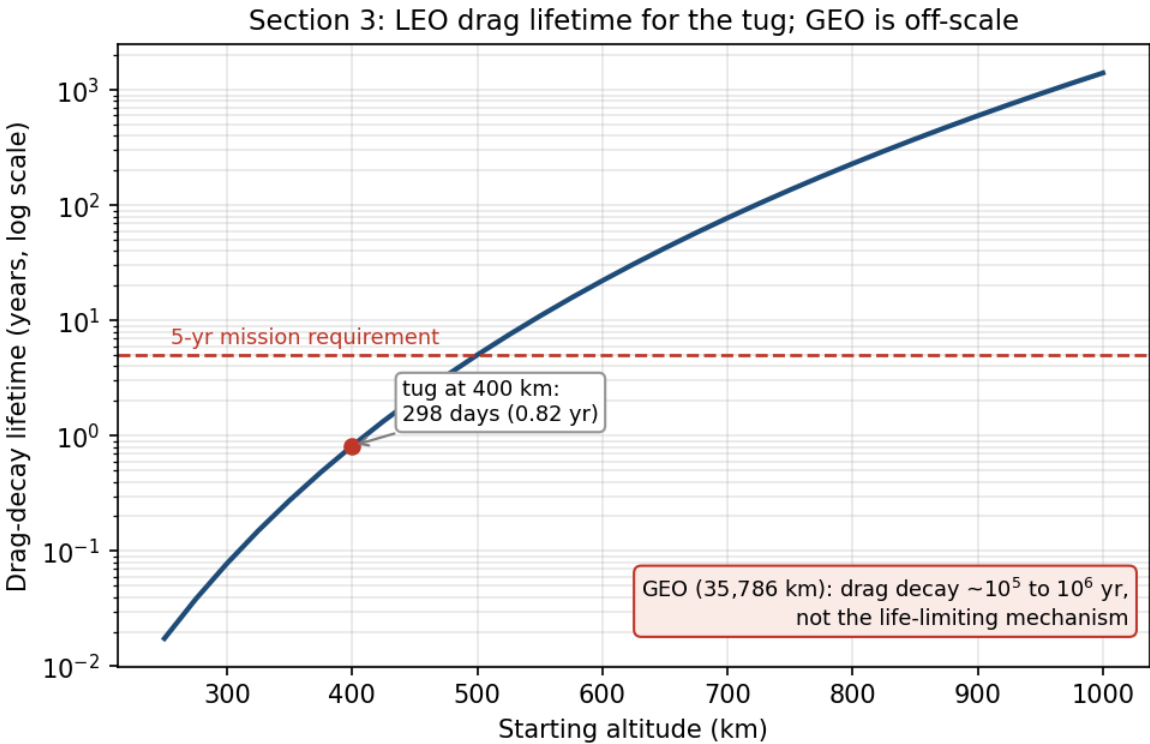


Figure 3: With the Homework 2 neutral-density model, LEO lifetime rises steeply with altitude; at GEO drag is negligible.

A LEO tug would have to start above roughly 500 km just to reach the five-year line. These figures are point estimates at a single density profile; because thermospheric density swells at solar maximum, the actual lifetime at a given altitude can shift by a factor of two to three over a solar cycle [2], [3], so the 400 km result of about 0.8 yr should be read as an order-of-magnitude value.

What actually evolves at GEO. Without stationkeeping the satellite does not deorbit, it drifts out of its slot. Luni-solar gravity grows the inclination at about 0.75 to 0.95 degrees per year, toward roughly 15° over 26.5 years, and this is the analemma that appears in the ground track of Section 12.1. Triaxiality drifts the longitude toward the stable points near 75° E and 105° W, and solar radiation pressure drives a small eccentricity oscillation [2], [3]. Holding the slot costs about 45 to 55 m/s per year north-south, which is the dominant term, plus about 2 to 4 m/s per year east-west [2], [3]. The result is Eq. (3):

Drag lifetime $\gg$ 5 yr (effectively unlimited); the real driver is ~ 50 m/s/yr N-S stationkeeping.

(3)

Assumptions: the wet mass and ram area above, the Homework 2 density fit for the LEO contrast, a standard GEO density of about 10^{-15} kg/m³, and standard luni-solar and triaxiality rates [2], [3]. MESA's life is therefore set by propellant and by wear-out, not by decay, and Section 7.4 turns that 50 m/s per year into a closed propellant budget.

4. The Sun-Earth System and Risks at GEO and LEO

Section 3 committed the vehicle to GEO. This section is the environmental case behind that commitment: what the Sun emits, what Earth's field does with it, and what the resulting hazard set looks like at the two candidate altitudes. The two regimes analyzed are **GEO** and **LEO**, which bracket the trade.

4.1 Solar emissions, quantified

The Sun drives the environment through four channels [4], [5]:

- **Electromagnetic radiation (photons):** the solar constant is about 1,361 W/m² at 1 AU [5], spanning X-ray and extreme-ultraviolet through visible to infrared. The EUV and X-ray end heats and ionizes the upper atmosphere.
- **Solar wind (charged particles):** a continuous plasma of protons and electrons at ~400 to 800 km/s and ~5 to 10 particles/cm³ [5], which pressurizes and shapes the magnetosphere.
- **Solar flares:** sudden X-ray bursts classed A, B, C, M, and X; M- and X-class flares cause sudden ionospheric disturbances and radio blackouts within minutes.
- **CMEs and SEPs (energetic particles and radiation):** coronal mass ejections hurl billions of tons of magnetized plasma that drive geomagnetic storms, and solar energetic particle events accelerate protons to tens or hundreds of MeV [5], arriving minutes to hours later as a penetrating radiation hazard on top of the always-present galactic cosmic ray background.

4.2 Earth's magnetic field and the radiation belts

Earth's magnetic field is roughly dipolar, about 30 μ T at the equator rising to about 60 μ T at the poles [4]. It carves out the magnetosphere, deflecting most of the solar wind, and traps charged particles in the Van Allen belts: an inner proton belt from roughly 1,000 to 6,000 km and an outer MeV-electron belt from roughly 13,000 to 60,000 km. This geometry is what drives the orbit tradeoff. LEO sits mostly beneath the belts and inside the protective field, while GEO at 35,786 km sits in the outer electron belt near the magnetosphere's edge, where strong CMEs can push the magnetopause inside GEO and expose satellites directly to shocked solar wind [5].

4.3 Risks at GEO

The leading cause of GEO anomalies is surface charging. GEO sits in the hot plasma sheet, and during substorms keV electrons charge exterior surfaces to kilovolt potentials. Because materials charge at different rates, the resulting differential charging drives electrostatic discharge into the electronics [4], [6]. Galaxy 15 is the canonical case: an ESD tied to disturbed space weather in 2010 latched its command unit, leaving it a powered but uncommandable "zombiesat" for eight months [7]. Section 9 quantifies the floating potential and the mitigations. MeV "killer electrons" from the outer belt compound this, penetrating the structure to charge internal dielectrics and discharge into buried circuits, a deep-dielectric mechanism that surface grounding cannot reach [6].

Energetic-particle radiation is the second major hazard. SEP protons and trapped electrons reach the vehicle with little natural shielding, so total ionizing dose accumulates across a mission of five years or more, while SEP and cosmic ray strikes produce single-event upsets and latch-ups. Section 10 sizes the shielding and the part class against this.

Solar radiation pressure rounds out the mechanical environment. Photon momentum is only about 4.5 μ N/m² at 1 AU, negligible as a drag force, but across MESA's array moment arm it is both an orbital perturbation, appearing as the eccentricity oscillation noted in Section 3.3, and the dominant attitude disturbance torque at 93.5 percent of the worst-case total in Section 11.

Two further effects shape the design. Eclipse seasons drive deep thermal cycling and solar ultraviolet degrades optical coatings [5], both addressed in Section 8, while charging upsets and ionospheric effects degrade the downlink, covered in Section 5. For MESA the charging risk is doubled, because an ESD during docking could damage the tug and the client at once.

4.4 Risks at LEO

LEO trades this hazard set for a different one. The South Atlantic Anomaly is the sharpest feature: the inner proton belt dips to LEO altitudes there, spiking dose and upset rates on every pass [5], [8]. Atmospheric drag dominates and couples strongly to solar activity, since EUV and X-ray heating expands the thermosphere so density and drag rise at solar maximum [4], [5]. Section 3.3 quantifies exactly this for the MESA ballistic coefficient.

Atomic oxygen is the distinctive LEO material threat. Solar ultraviolet dissociates molecular oxygen, and the resulting atomic oxygen at about 5 eV erodes polymers such as Kapton on ram surfaces at roughly 10²⁰ to 10²¹ atoms/cm² per year near 400 km [8]. High-inclination orbits add polar SEP access and auroral charging [5]. The operational consequences compound: solar activity perturbs orbit and pointing, ionospheric scintillation degrades GPS and downlinks, and SEP events add detector and star-tracker noise.

4.5 The comparison that decided the orbit

Table 2: GEO versus LEO environmental hazards for the servicing mission.

Hazard	GEO	LEO
Surface / deep-dielectric charging and ESD	Severe (plasma sheet, killer electrons)	Milder, mainly auroral
Van Allen belt / energetic-particle dose	High (in the outer belt, little shielding)	Lower, spikes in the SAA and at poles
Atmospheric drag	Negligible	Dominant below ~600 km

Hazard	GEO	LEO
Atomic oxygen erosion	None	Significant on ram surfaces
Solar radiation pressure	Dominant attitude disturbance on large arrays	Negligible versus drag
Debris flux	Low	High
Thermal cycling	Deep at eclipse seasons	About sixteen cycles per day

Table 2 says that the GEO column is dominated by electrical effects and the LEO column by mechanical and chemical ones. Electrical effects are addressed with coatings, bonding, part selection, and operational rules, all of which are cheap in mass. Mechanical and chemical degradation is addressed with propellant and with material loss that never stops. That is the deeper reason GEO is the right regime for a vehicle expected to work for five years and then keep going.

5. Space Weather: Monitoring and Downlink Impact

The hazards in Section 4 are not steady. They surge with solar activity, which makes forecasting them an operational requirement rather than a convenience for a vehicle that must decide when it is safe to approach another satellite.

What it is. Space weather is the set of conditions on the Sun and in the solar wind, magnetosphere, ionosphere, and thermosphere that can affect space- and ground-based technology and endanger operations [9].

Monitoring and research. The U.S. operational center is NOAA's Space Weather Prediction Center, which issues the watches, warnings, and alerts that operators act on [9]. It is fed by a layered observing system, and what matters operationally is which asset gives which warning and how far ahead.

At GEO, the GOES satellites carry X-ray sensors that detect flares as they happen and set the flare class, energetic particle sensors that measure the proton flux driving the NOAA S-scale radiation storm levels, and magnetometers that record the local field distortion during storms. Because X-rays travel at the speed of light, a GOES flare detection arrives about eight minutes after the event on the Sun, which is warning of the radio blackout but not of the particles that follow.

At the L1 Lagrange point, roughly 1.5 million km sunward, DSCOVR and NASA's ACE sample the solar wind speed, density, and embedded magnetic field directly. Because they sit upstream, they see a CME shock front before it reaches Earth and provide the single most actionable number in the system: approximately fifteen to sixty minutes of lead time, depending on shock speed, between the L1 measurement and the geomagnetic response at Earth. That window is what allows a docking hold to be commanded before conditions turn.

Solar imaging provides the longer horizon. The Solar Dynamics Observatory and SOHO image the corona and photosphere continuously, so an erupting CME can be identified and its arrival time estimated one to three days ahead, which is long enough to reschedule a servicing operation rather than merely abort one. Ground networks close the loop. Magnetometer chains yield the planetary Kp index and the Dst index that tracks ring-current intensity, neutron monitors detect ground-level enhancements from the most energetic solar protons, and ionosondes profile the ionosphere that the downlink must cross.

On the research side, physics-based magnetohydrodynamic models of CME propagation now drive the operational arrival-time forecasts, and empirical specification models of the trapped radiation belts and the thermosphere supply the environment definitions used for design work of exactly the kind in Sections 8 through 11. The practical limitation is that arrival-time forecasts still carry uncertainty measured in hours, which is why the design cannot rely on forecasting alone and must also tolerate the environment.

Why the customer wants it. MESA docks next to a live client, so a charging event during proximity operations risks ESD damage to both vehicles, and forecasts let operators hold docking during disturbed conditions [4], [6]. An SEP warning lets the tug safe its avionics or delay a burn, and routine nowcasts feed drift and downlink-reliability planning. For this mission the value of the SWPC feed is concrete and it is a contractual point rather than a nicety: it converts an uncontrolled risk during the most delicate phase of the mission into a scheduling constraint, which is the gate drawn in Figure 1.

Downlink impacts from GEO to ground. MESA's telemetry, tracking, and command link from a fixed GEO longitude to a fixed ground station is geometrically simple, but space weather still degrades it [10]. Ionospheric scintillation is the most common effect, because the slant path crosses the ionosphere and disturbed conditions cause amplitude and phase fading, worst at equatorial and auroral latitudes and at lower frequencies. Twice a year near the equinoxes, solar radio frequency interference produces a sun outage: the Sun passes directly behind the GEO satellite as seen from the ground station, and solar radio noise raises the receiver noise temperature enough to black out the link for several minutes a day across several days, with solar radio bursts adding sporadic interference on top. Changing total electron content rotates the signal polarization through Faraday rotation and adds group delay. Finally, the link can fail at the source, because a charging- or SEP-induced upset on the satellite transmitter interrupts the downlink regardless of propagation conditions [6]. The GEO downlink is therefore robust in normal conditions but must be scheduled around predictable sun outages and monitored during storms, which is exactly why the customer wants an SWPC feed.

6. Vacuum Testing

Space weather can be scheduled around. The vacuum environment cannot, because the vehicle sits in it continuously for the whole mission, which is why the customer's proposal to delete vacuum testing to save cost deserves a direct answer.

For a GEO servicing tug whose optical RPO sensors and docking mechanisms are the mission, deleting that campaign is a false economy, and **the recommendation is a full thermal-vacuum and thermal-balance test campaign with a pre-ship bakeout.** The case rests on four effects.

- **Outgassing and molecular contamination.** In vacuum, adsorbed water, solvents, and plasticizers evaporate and redeposit on cold surfaces such as optics, radiators, solar cells, and sensors. Materials are screened to ASTM E595 limits of total mass loss below 1.0% and collected volatile condensable material below 0.10% [11]. On MESA a film on the docking cameras or LIDAR blurs the sensors the mission depends on, and deposits on radiators and cells cut thermal and power performance by several percent [5], [11]. A pre-flight bakeout drives the volatiles off before launch.
- **Thermal-vacuum cycling.** Space rejects heat only by radiation, and GEO eclipse seasons swing components from full sun to shadow. TVAC verifies the thermal design and the workmanship behind it, including solder joints, connectors, and bondlines, across the flight range plus margin, and screens infant-mortality defects [12]. Section 8 puts that steady-state range at a swing of 49 K, and Section 12.3 shows which components actually see it.
- **Cold welding and mechanism survivability.** Bare metal contacts can cold-weld in vacuum, and liquid lubricants evaporate. MESA's docking mechanism, robotic arm, and deployables must be qualified in vacuum with space-rated dry lubricants and materials [13]. This is the single failure mode that would end the mission with the vehicle otherwise healthy, because a capture mechanism that will not release strands MESA on its first client.
- **Multipaction and corona.** High-power radio frequency components in the communications chain can suffer multipaction discharge in vacuum and must be tested for it [5].

The cost argument. Skipping TVAC to save a small fraction of a \$100M program risks the entire asset plus the client it is servicing. MESA cannot repair itself on orbit, so an undetected workmanship or contamination failure is mission-ending: one hazed docking sensor could abort every rendezvous, and a few percent of lost array or radiator performance compounds over a five-year life. The \$9M integration and test line in Table 8 buys down a \$100M loss plus the client's asset alongside it. The calculus only flips for a low-cost, high-quantity CubeSat build where a single unit is expendable, and MESA is the opposite of that.

7. Vehicle Definition and System Budgets

Sections 2 through 6 established what MESA does and what it must survive. This section defines the vehicle that does it and shows that the design closes: mass, power, propellant, and cost each land inside their allocation with margin. Everything downstream draws its geometry from here.

7.1 Configuration

Table 3 defines the configuration. **Figure 4** shows the deployed vehicle and the resulting mass properties.

Table 3: MESA configuration. The wet mass is the program requirement; the geometry is sized here.

Parameter	Value	Note
Wet mass	2,000 kg	1,400 kg dry, 600 kg propellant
Bus envelope	1.8 x 1.8 x 3.5 m	z is the nadir and docking axis
Solar wings	2 x 5.12 m ² = 10.24 m ²	single-axis sun tracking, 60 kg each
Bus radiating area	31.68 m ²	total exterior
Bus sun-projected area	6.30 m ²	one large face, worst case
Illuminated area A_s	16.54 m ²	arrays plus bus, used for SRP and drag
I_x, I_y, I_z	3,279 / 2,452 / 1,893 kg·m ²	free flyer, computed in the script
Client envelope, assumed	2.0 x 2.0 x 3.0 m	representative GEO comsat, Appendix B
Client centre of mass, from MESA	3.0 m along $+z$	docking standoff, Appendix B
I_x mated with a 3,000 kg client	17,329 kg·m ²	5.3 times the free flyer
c_{ps} to c_m offset	0.25 m	arm and docking hardware are off-axis; Appendix B
Residual magnetic dipole D	5 A·m ²	assumed, typical for this class
Internal dissipation Q_{int}	1,200 W	orbit-average, from the Table 5 budget

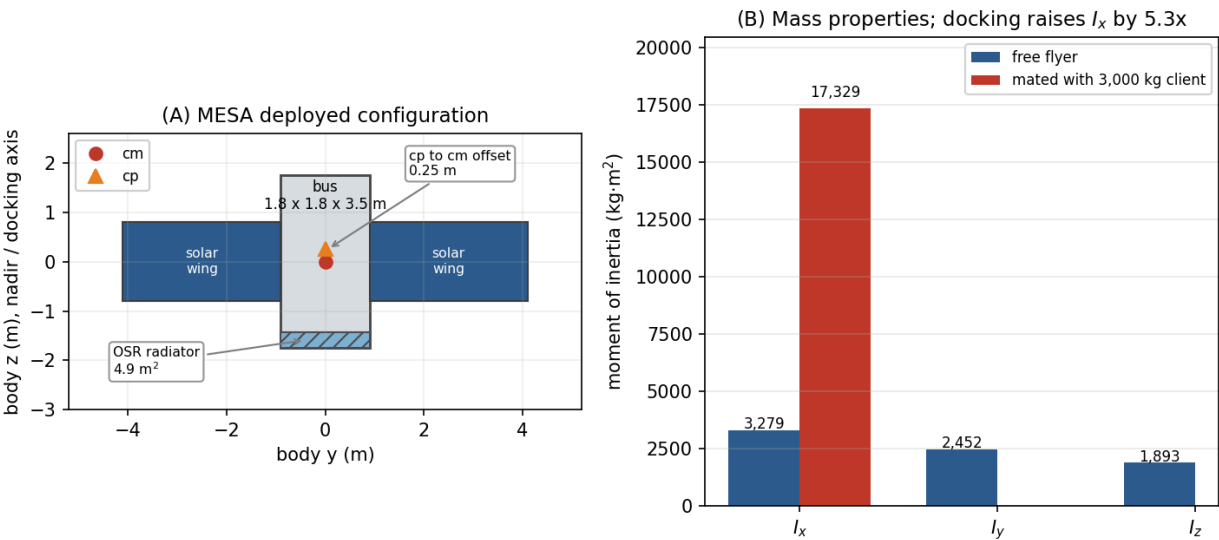


Figure 4: Sized MESA configuration and mass properties. The wheels must be sized for the mated stack, not the free flyer.

The mated inertia is the number that matters most in this table. Capturing a 3,000 kg client shifts the combined center of mass 1.80 m along the docking axis and raises I_x by a factor of 5.3, so the ADACS in Section 11 is sized for the mated stack rather than for the free flyer.

7.2 Mass budget

Table 4 closes the mass budget against the 2,000 kg wet requirement, and Figure 5(A) shows it graphically. Subsystem allocations are engineering estimates for this vehicle class except for the ADACS line, which is the computed result from Section 11.3, and the propellant line, which is the load Section 7.4 sizes.

Table 4: MESA mass budget.

Item	Mass (kg)	Basis
Structure and mechanisms	300	primary and secondary structure, deployables
Docking mechanism and robotic arm	180	capture latch, arm, servicing tooling
ADACS (wheels, sensors, RPO suite)	157	computed, Section 11.3
Propulsion, dry	165	xenon and hydrazine tanks, PPU, thrusters, lines
Power (arrays, battery, PCU, harness)	268	includes 120 kg of wings and a 38 kg battery
Thermal control	65	MLI, 4.9 m² OSR radiator, heaters, heat pipes
C&DH and avionics	75	includes 15 kg of 100 mil shielding, Section 10.2
Communications	40	TT&C transponders and antennas
Subsystem total	1,250	
Margin	150	10.7% of dry mass
Dry mass	1,400	
Propellant	600	520 kg xenon, 80 kg hydrazine
Wet mass	2,000	closes against the requirement

The heavy lines are the ones that make this a servicer rather than a bus. The docking mechanism and arm at 180 kg plus the 25 kg RPO sensor suite account for about fifteen percent of the dry mass on their own, and they are exactly what a conventional communications satellite of the same size does not carry.

7.3 Power budget

Table 5 gives the electrical load, and Figure 5(B) compares it against the array. Both the peak case, with every load energized, and the orbit-average case are shown, because the array is sized on the first and the thermal design is driven by the second.

Table 5: MESA electrical load.

Load	Power (W)
ADACS (wheels, sensors, electronics)	283
Avionics and C&DH	150
Communications (TT&C)	120
RPO sensors (LIDAR, cameras)	60
Thermal heaters (eclipse)	260
Electric stationkeeping thruster	600
Robotic arm and servicing payload	300
Peak load	1,773
Orbit-average load	1,200

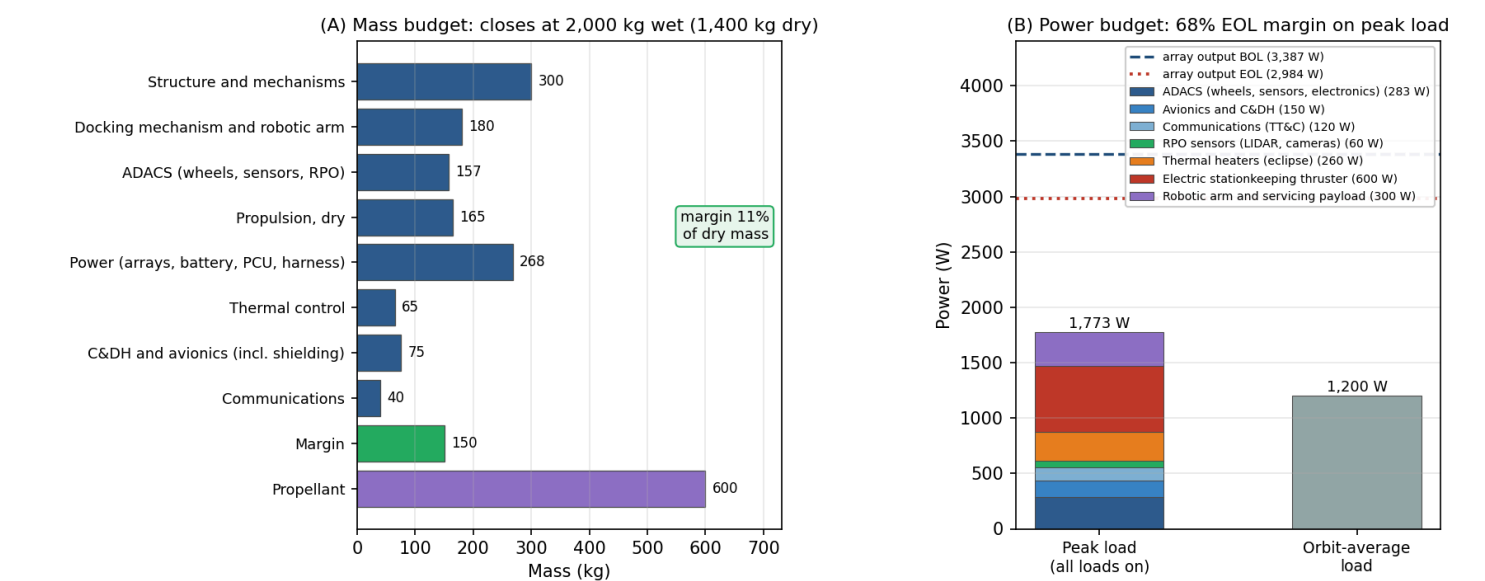


Figure 5: Both budgets close with margin. The array is sized by the end-of-life case, not the beginning-of-life case.

The array is 10.24 m² of triple-junction gallium arsenide at thirty percent beginning-of-life efficiency, with a 0.90 packing factor and a 0.90 hot-cell derate, giving 3,387 W at beginning of life and 2,984 W after five years of radiation damage. Section 12.2 derives both numbers and sizes the battery. The relevant budget result is that the array is sized on the end-of-life case and still carries 68 percent margin over peak load, which is what allows the peak case to be treated as genuinely available rather than as a paper number.

The orbit-average figure of 1,200 W is also the internal dissipation Q_{int} used in the thermal analysis, on the assumption that essentially all electrical power ends up as waste heat inside the bus.

7.4 Propulsion, delta-v, and propellant

MESA carries two propulsion systems, and the split is deliberate. A xenon Hall thruster in the 600 W class, delivering roughly 40 mN at 1,600 s specific impulse [3], does the slow, high-total-impulse work: stationkeeping the mated stack, tows to and from the graveyard, and longitude relocations. A hydrazine monopropellant system at 220 s does the work electric propulsion cannot: proximity operations, where thrust must be available on demand and in any direction, and momentum dumping.

Table 6 allocates the five-year delta-v against the baseline manifest of six client servicing cycles from Figure 1, and Figure 6(A) shows the same allocation graphically.

Table 6: Five-year delta-v budget.

Task	Delta-v (m/s)	Basis
Mated stationkeeping, four years on the stack	200	50 m/s/yr, Eq. (3)
Free-flight stationkeeping, one year	50	50 m/s/yr, Eq. (3)
Graveyard tows, six round trips	131	Hohmann, 10.88 m/s each way for +300 km
Longitude relocations, six	34	5.69 m/s per repositioning at 1°/day drift
RPO, approach and backout, six	48	8 m/s per client, approach plus a contingency abort

Task	Delta-v (m/s)	Basis
Total	463	

$$\Delta v_{5\text{ yr}} = 463\text{ m/s}$$

(4)

Converting that to propellant through the rocket equation, $m_p = m_0 \left(1 - e^{-\Delta v / I_{sp} g_0}\right)$, in which m_p is the propellant consumed (kg), m_0 the mass at the start of the maneuver (kg), Δv the velocity increment (m/s), I_{sp} the specific impulse (s), and g_0 standard gravity (9.807 m/s²), applied at the mass each task is actually flown at, gives **Table 7**. Stationkeeping and towing are flown at the 5,000 kg mated mass; relocations and proximity operations are flown at the 2,000 kg free-flyer mass.

Table 7: Propellant consumption against the 600 kg load, split 520 kg xenon and 80 kg hydrazine.

Task group	System	I_{sp} (s)	Flown at (kg)	Propellant (kg)
Mated stationkeeping and tows	Xenon Hall	1,600	5,000	104.8
Free-flight stationkeeping and relocations	Xenon Hall	1,600	2,000	10.7
Xenon consumed of 520 kg loaded				115.5 (78% reserve)
RPO, approach and backout	Hydrazine	220	2,000	44.0
Momentum dumping, 101 dumps	Hydrazine	220	2,000	5.2
Hydrazine consumed of 80 kg loaded				49.2 (38% reserve)
Total consumed of 600 kg loaded				165

$$m_{prop, used} = 165\text{ kg of 600 kg loaded; capacity} \approx 10\text{ client cycles, set by the hydrazine tank}$$

(5)

The two tanks are sized separately, so the capacity is set by whichever runs dry first rather than by the combined 600 kg. **Figure 6(B)** plots each tank as a fraction consumed against the number of client cycles completed, and the crossing is unambiguous: xenon would support 28 cycles, hydrazine supports 10.2, so hydrazine is the binding sub-budget. If the customer wants more cycles under a contract extension, the fix is a larger monopropellant tank, not more xenon.

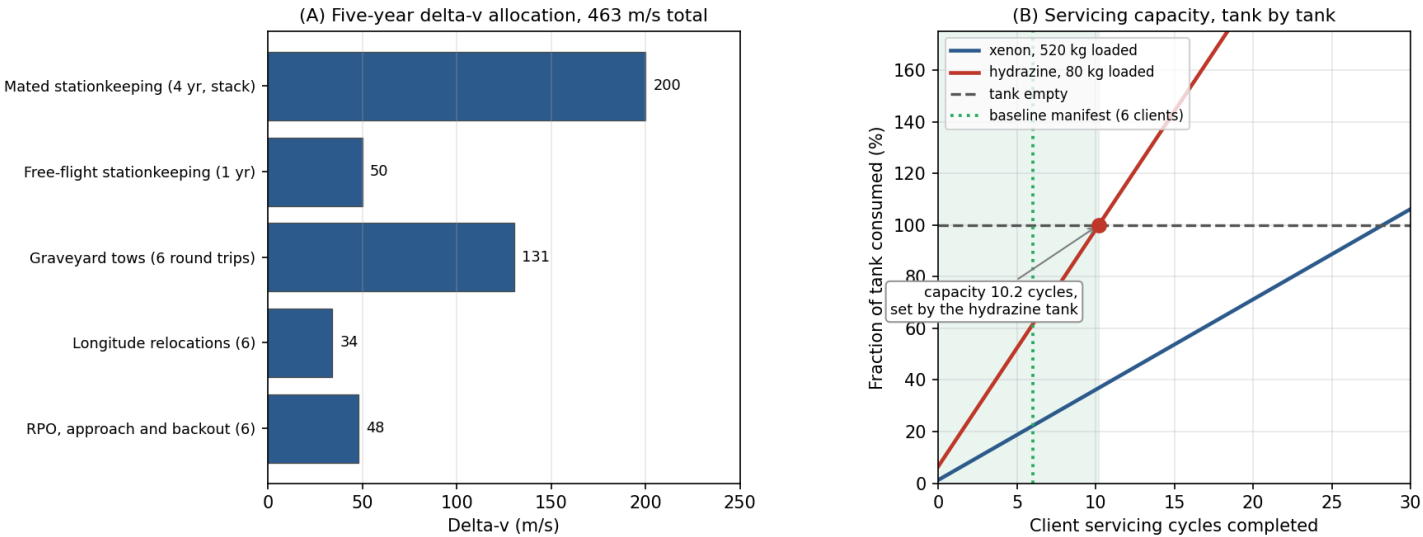


Figure 6: Propellant sizes the servicing capacity, not the calendar life. The baseline manifest leaves capacity in reserve.

Two further things follow, and the second one refines a conclusion carried through Milestone 2. First, the electric thruster is comfortably able to do the job: holding a 5,000 kg stack against 50 m/s per year needs 685 N-s per day, which is 4.8 hours of firing per day at 40 mN, well inside a normal electric-propulsion duty cycle. Second, **propellant sizes the servicing capacity, not the calendar life**. Milestone 2 concluded that stationkeeping propellant was the binding life limit, and with a chemical system that would have been right. With electric propulsion at 1,600 s the load carries roughly ten client cycles against a baseline manifest of six, so what actually bounds a five-year contract is wear-out: reaction wheel bearing life, docking mechanism cycles, and battery charge-discharge cycles. That is a better problem to have, because it is addressed with qualification testing rather than with tankage.

7.5 Cost

The \$100M cap is a design-to-cost constraint, so **Table 8** allocates it rather than estimating it bottom-up. The allocation reflects where this vehicle is genuinely different from a communications bus: the servicing hardware and the sensor suite carry a third of the program, and the radiation-hardened parts in Section 10 are expensive in a way commercial equivalents are not.

Table 8: Design-to-cost allocation, excluding launch.

Element	Cost (\$M)
ADACS and RPO sensor suite	18
Docking mechanism and robotic arm	15
Avionics, C&DH, communications (RHA category R parts)	12
Bus structure and mechanisms	12
Power subsystem	10
Integration, test, and the TVAC campaign of Section 6	9
Propulsion	8
Program management and systems engineering	7
Thermal control	4
Program reserve	5
Total	100

Two caveats belong on this table rather than in a footnote. Launch is excluded and is assumed customer-furnished or procured separately, which is normal for a servicing vehicle whose ride is often a shared GTO slot. And the allocation is aggressive: a 2,000 kg servicer at \$100M is only achievable on a heritage bus with a largely qualified parts list, which is precisely why the design follows proven MEV practice [1] rather than inventing an architecture.

The value argument closes the case. A GEO communications satellite generates revenue measured in tens of millions of dollars per year, and MESA extends six of them by roughly a year each within one contract while freeing the operational slots it clears. The asset it protects on any single mission is worth more than the tug.

8. Thermal Control System

Section 7 fixed the geometry and the 1,200 W of internal dissipation. At GEO the vehicle can shed that heat only by radiation, so the thermal design reduces to a single question: how much radiator area, and what closes the cold case.

8.1 Equilibrium temperatures in sun and eclipse

Treating the bus as isothermal and balancing absorbed solar flux, absorbed Earth flux, and internal dissipation against re-radiation [14] gives Eq. (6):

$$\alpha F_s A_{proj} + Q_{Earth} + Q_{int} = \sigma (\varepsilon_{MLI} A_{MLI} + \varepsilon_{OSR} A_{rad}) T^4$$

(6)

where α is the solar absorptivity of the outer surface, F_s the solar constant (1,361 W/m²), A_{proj} the sun-projected area (m²), Q_{Earth} the absorbed Earth infrared plus albedo load (W), Q_{int} the internal dissipation (1,200 W), σ the Stefan-Boltzmann constant (5.670×10^{-8} W/m²K⁴), ε_{MLI} and ε_{OSR} the infrared emissivities of the blanket and the radiator, A_{MLI} and A_{rad} their areas (m²), and T the equilibrium temperature (K). The left side is every watt absorbed or generated; the right side is every watt radiated away.

The Earth terms are negligible at this altitude. Scaled by the $(R_E/r)^2$ view factor, Earth infrared is 5.42 W/m² and albedo 9.34 W/m², together 4.77 W against roughly 2,400 W of solar and internal load, or two tenths of one percent. The script carries them; the hand calculation drops them.

Setting the sunlit case to 310 K sizes the radiator at **4.9 m² of optical solar reflector**. Holding that area fixed and re-solving Eq. (6) with and without the solar term gives the two required temperatures:

$T_{sun} = 309.5 \text{ K} = +36.3 \text{ }^\circ\text{C}$

$T_{eclipse} = 260.1 \text{ K} = -13.1 \text{ }^\circ\text{C}$

(7)

That is a steady-state swing of 49.4 K, and the shape is the expected one: the sunlit case sits inside a normal electronics band and the eclipse case falls below it, because radiators are sized by the hot case and heaters close the cold case. Holding an isothermal bus at 0 °C against the eclipse case needs **260 W** of heater power, which is the line carried in Table 5. **Figure 7** shows the sizing trade.

Section 8: MESA isothermal-bus thermal balance at GEO

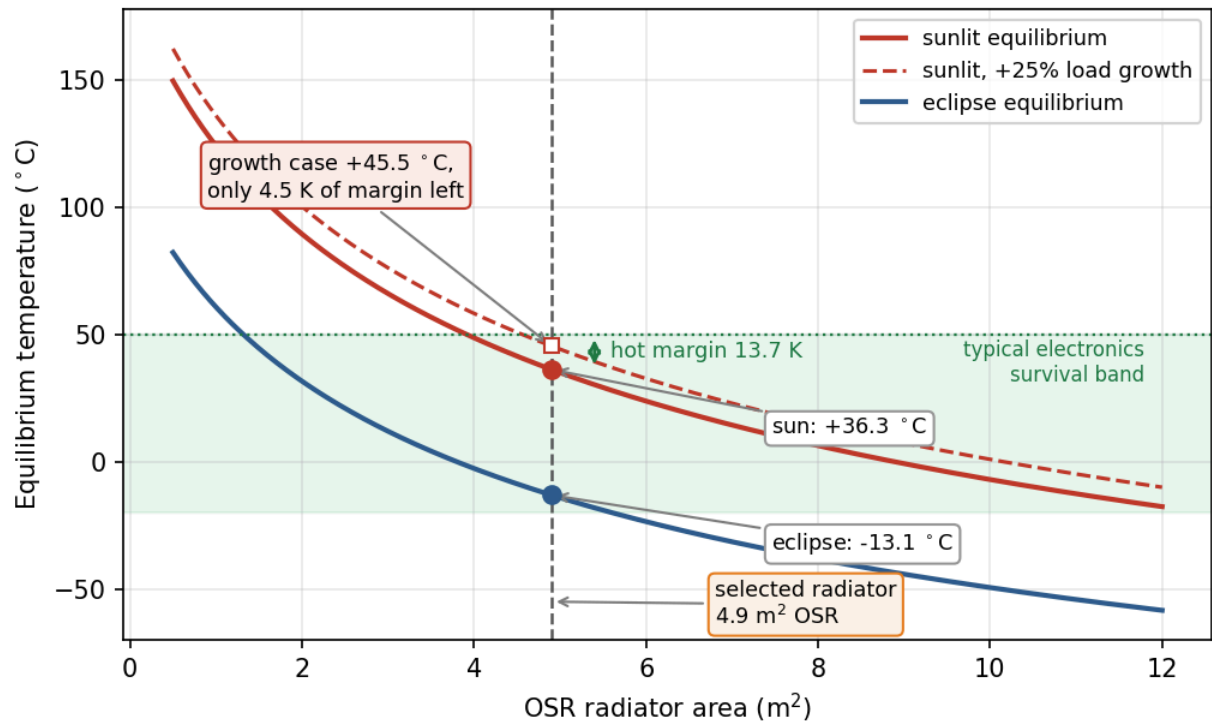


Figure 7: Sizing the radiator is a trade between the sunlit and eclipse cases. The load-growth case is what sets the hot margin, and heaters close the eclipse gap.

One important qualification, and it is the reason Section 12.3 exists: Eq. (7) is a pair of steady-state bounds, not a prediction of what the bus does during a 69 min eclipse. The transient simulation shows the bus itself moves only 3.9 K, and that the 260 W of heater power is genuinely needed by the low-mass outboard zones rather than by the bus. The steady-state numbers are still the right ones to design the radiator against, because the sunlit case is reached and held for most of the year.

8.2 Thermal design margin

Equilibrium temperatures on their own do not show whether a design is safe, only where it sits. Table 9 puts the predictions against the component limits and states the margin in each case. The limits are the conventional electronics band, +50 °C hot and -20 °C cold, which is the band shaded in Figure 7.

Table 9: Thermal margin assessment at the 4.9 m² baseline radiator.

Case	Predicted	Limit	Margin
Sunlit, nominal 1,200 W internal load	+36.3 °C	+50 °C	13.7 K
Eclipse, steady-state bound	-13.1 °C	-20 °C	6.9 K
Eclipse, transient prediction (Section 12.3)	+32.4 °C	-20 °C	52.4 K
Sunlit, with a 25% internal load growth allowance	+45.5 °C	+50 °C	4.5 K

Three things follow from that table.

The hot case is the binding one, and it is the case the radiator is sized against. The nominal 13.7 K clears the 11 K uncertainty allowance normally carried on an analysis-only prediction before thermal balance test, but only just, and the growth case does not clear it at all. Load growth is therefore controlled as a requirement rather than absorbed later: the 1,200 W orbit-average dissipation in Table 3 is a not-to-exceed number levied on the subsystems, not a best estimate.

The cold case is not as tight as the steady-state bound suggests. The -13.1 °C figure is where an isothermal bus would settle given unlimited time in shadow. Section 12.3 integrates the actual 69.4 min eclipse and gets +32.4 °C, because the vehicle's thermal time constant is roughly ten times the eclipse. The cold margin that matters is the component-level one on the outboard zones, and that is what the heaters own.

Oversizing the radiator is a trade, not a free improvement. Growing the radiator from 4.9 to 5.5 m² would take the sunlit case to +29.2 °C and raise the hot margin to 20.8 K, which looks attractive until the cold side is checked: the same change drives the isothermal eclipse bound from -13.1 °C to -19.0 °C and raises the heater power needed to hold the bus at 0 °C from 260 W to 402 W. That 142 W is a direct charge against the power budget in Table 5 and against the battery in Eq. (14). The 4.9 m² baseline is selected because it is the smallest area that clears the hot limit with the uncertainty allowance intact, which leaves the cold case to the heaters, where it is cheaper to solve.

The residual risk is concentrated in the growth case, and it is retired the same way every thermal program retires it: by correlating the model against hardware in the thermal balance test recommended in Section 8.4, and by holding radiator panel growth as a mass and area reserve until that correlation exists.

8.3 Recommended thermal control system and why

The design is passive-dominant, which is the right choice for a vehicle with a steady internal load and no agile thermal requirement.

- **MLI blankets** over the bus, with a silverized-Teflon outer layer at $\alpha = 0.14$ and effective $\varepsilon = 0.03$. The low absorptivity keeps the sunlit case cool, and the low effective emittance is what makes the eclipse case survivable at all. The outer layer is ITO-coated for charge control, which is a Section 9 requirement driving a Section 8 part choice.
- **OSR radiators**, 4.9 m², mounted on the anti-sun face so they never see direct sun. Second-surface mirrors hold a low α/ε ratio far better than white paint across five years of ultraviolet exposure, which matters on a vehicle that cannot be recoated.
- **Heaters with redundant thermostats**, 260 W total, zoned on the battery, the propellant lines and tanks, the docking mechanism, and the arm joints rather than on the whole bus. Section 12.3 shows why that zoning is not just good practice but the actual requirement: the propellant lines reach -0.2 °C in eclipse without a heater, below hydrazine's 2 °C freezing point, while the bus barely moves.
- **Heat pipes and doublers** running from the avionics and battery to the radiator, so the dissipating boxes are coupled to the rejection path instead of relying on the structure to carry heat.
- **Thermal isolation at the docking interface.** A client that has been powered down is cold, and a conductive path into a captured client would pull MESA's bus down with it, so the capture mechanism uses low-conductance standoffs. This is a servicer-specific requirement with no analogue on a conventional satellite.

8.4 Ground testing recommendation

The recommendation is the **full TVAC and thermal-balance campaign from Section 6**, and the numbers above are the reason. A predicted swing of 49 K and a predicted 3.9 K orbital excursion both rest on assumed values of α , ε , and Q_{int} , and thermal balance testing is how those assumptions get correlated to hardware before flight rather than after launch. The same vacuum campaign is what qualifies the deployment and capture mechanisms against cold welding, a vacuum failure mode that no ambient test will reveal [13].

If the customer deletes ground testing anyway, the thermal control system absorbs that uncertainty in margin and material choice: a design band widened to roughly ± 25 K, coatings restricted to flight-proven silverized Teflon and OSR with published beginning- and end-of-life properties, the radiator oversized by about thirty percent against an under-predicted internal load, and heater strings fully cross-strapped, since an uncorrelated cold case ends the mission. That extra area, power, and harness costs mass on every flight unit. Testing once is cheaper.

9. Plasma Environment

Section 4.3 identified surface charging as the leading cause of GEO anomalies. This section quantifies it for MESA and works the design response, including the one charging failure mode that is unique to a servicing vehicle.

9.1 Risks

The vehicle is electrically isolated from the plasma described in Section 4.3, so it floats to whatever potential balances the currents reaching it. That potential is worth deriving rather than quoting, because every mitigation below is sized against it.

Electrons and ions in the plasma sheet share roughly the same temperature, but an electron is about 1.8×10^3 times lighter than a proton and therefore moves about 43 times faster at that temperature. An uncharged surface would collect electrons far faster than ions, so it charges negative, and it keeps charging until the growing negative potential repels enough of the incoming electron flux to match the ion flux it still collects. Setting those two currents equal and solving for the potential leaves a result proportional to the electron temperature, with a coefficient set by the electron-to-ion mass ratio and by the current-collection model assumed. The value this course uses for a hydrogen plasma sheet is 2.50, giving Eq. (8) [15]:

$$V = -2.50 \frac{k_B T_e}{q_e} \approx -2.16 \text{ kV} \quad (8)$$

where V is the floating potential of the vehicle relative to the surrounding plasma (V), k_B the Boltzmann constant (1.381×10^{-23} J/K), T_e the plasma sheet electron temperature (K), and q_e the elementary charge (1.602×10^{-19} C).

The arithmetic is worth showing, because the answer is only as good as the electron temperature that goes into it. Taking the substorm plasma sheet at $T_e = 10^7$ K, which is the environment definition this course uses for GEO charging [15],

$$\frac{k_B T_e}{q_e} = \frac{(1.381 \times 10^{-23})(10^7)}{1.602 \times 10^{-19}} = 862 \text{ V}, \quad V = -2.50 \times 862 = -2,155 \text{ V} \quad (9)$$

which rounds to the -2.16 kV quoted above. Two checks on that number. The ratio $k_B T_e / q_e$ is just the electron temperature expressed in volts, and 10^7 K is about 862 eV, so a kilovolt-scale answer is the right order of magnitude for a keV plasma. And the potential scales linearly with T_e , so a quieter plasma sheet at 10^6 K would float the vehicle at only -216 V, which is why charging is a substorm problem rather than a continuous one.

The absolute potential is not what breaks hardware. A well-bonded conductive vehicle can sit a couple of kilovolts below its environment and function normally. The danger is differential charging: coverglass, Kapton, and metal structure charge to different potentials, and once the gap between them exceeds the breakdown threshold the result is an arc [6], [15]. That arc is the leading cause of GEO anomalies and the mechanism behind the Galaxy 15 loss of command for eight months [7]. Deep-dielectric charging compounds the problem, because MeV outer-belt electrons penetrate the skin entirely and deposit charge inside cable dielectrics and circuit boards, which then discharge into buried signal lines [6]. Surface grounding does nothing for this mechanism; only shielding and bulk conductivity help.

What turns a charging event into a mission-ending one is the coupling that follows the arc. The transient couples into the harness and avionics, which is exactly how a surface effect becomes an uncommandable vehicle [6]. On the solar array a primary arc between adjacent strings can be sustained by the array’s own current, permanently shorting a string section. A negatively biased vehicle also attracts its own outgassed contaminants back onto cold surfaces, which lands directly on the RPO optics this mission depends on [5].

The mission-unique risk is the docking interface. MESA and its client float independently and reach different potentials, because they have different surface materials, different areas, and different illumination histories. At the moment of capture the two vehicles are bonded through the docking mechanism, and the potential difference equalizes through whatever path is available. If that path is the capture latch or a signal umbilical, the discharge damages the mechanism or the avionics on both vehicles, and MESA has just destroyed the asset it was sent to save. No conventional satellite carries this failure mode, because no conventional satellite deliberately touches another one.

9.2 Mitigations

The baseline is a fully conductive, bonded exterior per NASA-HDBK-4002A [6]: ITO-coated coverglass, an ITO-coated MLI outer layer, conductive paint on the remaining surfaces, and grounding straps across every hinge and deployable, all tied to a common structure ground. That architecture is enforced through single-point grounding per NASA-HDBK-4001 [16], so there is one defined return path and no floating islands left to charge differentially. For the deep-dielectric mechanism, which grounding cannot reach, the answer is the 100 mils of aluminum equivalent from Section 10 combined with slightly conductive dielectrics, so deposited charge bleeds off faster than it accumulates. Harness entering the avionics carries EMI filtering and transient suppression, and the array uses arc-tolerant string isolation diodes.

The docking-interface risk gets a dedicated fix. MESA carries a plasma contactor that emits a low-energy electron current, clamping the vehicle near plasma potential. The operational rule is to run the contactor and drive both vehicles toward a common potential **before** mechanical contact, so equalization happens through a controlled emissive path rather than through the capture latch. Layered on top of the hardware, docking windows are scheduled against the SWPC feed from Section 5, and proximity operations hold during substorm conditions and Kp excursions. This is the hold gate drawn in Figure 1, and it is the reason the space weather feed is a contractual requirement rather than a convenience.

9.3 Impact on the other subsystems

The mitigations above are not free to the rest of the vehicle. **Table 10** works through what each one costs.

Table 10: Impact of the plasma mitigations on the rest of the vehicle.

Subsystem	Impact of the plasma mitigations
Thermal	ITO on the MLI and coverglass raises α and shifts α/ε , feeding back into the Section 8 radiator sizing. Coatings must be selected for charge control and optical properties at once.
Power	The plasma contactor draws roughly 30 W when active and adds about 10 kg with its gas supply, both carried inside the Table 4 and Table 5 allocations. Array string isolation costs a small amount of conversion efficiency.
C&DH and avionics	EMI filtering and transient suppression add mass and part count, and the software must tolerate and recover from arc-induced upsets rather than assume clean power.
Structures	Grounding straps across every hinge, deployable, and the docking mechanism, plus conductive-path continuity verification as a manufacturing requirement.
ADACS	Arc transients raise the noise floor on star trackers and sun sensors, so the estimator must reject dropouts rather than track them.
Operations	Docking is no longer schedulable on demand but gated on space weather, which lengthens the servicing timeline per client and is built into the 242-day cycle of Figure 1.

The honest summary is that plasma survivability is cheap in mass and expensive in discipline. The hardware fixes amount to coatings, straps, and one small contactor. What they actually cost is a grounding architecture enforced across every box and every interface, plus an operational constraint on when the mission’s central activity is allowed to happen.

10. Radiation Environment

Plasma acts on the vehicle’s surfaces. Radiation passes through them, so the design response shifts from coatings and bonding to part selection and shielding.

10.1 Risks

GEO sits inside the outer electron belt, outside most of the geomagnetic shielding, for the entire mission [17]. The three sources are trapped MeV electrons and their bremsstrahlung, solar energetic protons during events, and the galactic cosmic ray background.

Total ionizing dose is the cumulative threat. Charge trapped in gate oxides shifts threshold voltages and increases leakage until parts fail, and it accumulates continuously across five years with no recovery. Single-event effects are the acute counterpart: a single heavy ion or high-energy proton deposits enough charge in a sensitive volume to flip a bit, latch a parasitic structure into a high-current state, or rupture a power device gate [17]. Latch-up and gate rupture are destructive, while upsets are recoverable if the design anticipates them. Susceptibility varies sharply by technology, running from radiation-hardened CMOS on sapphire or insulator at the safe end to NMOS dynamic

memory at the vulnerable end [14]. Displacement damage is the third mechanism, in which non-ionizing energy loss knocks atoms out of the lattice and degrades minority carrier lifetime; on the solar array this is the dominant cause of the power decay that sets end-of-life output.

Solar particle events concentrate all of this into hours. A severe or extreme storm can cause memory device problems, star-tracker interference severe enough to lose orientation, and permanent solar-panel degradation from a single event [17]. For a vehicle performing precision proximity operations, losing the star trackers mid-approach is the acute risk, and it is what drives the operational rules below.

Dose estimate. The design assumes **5 krad(Si) per year behind 100 mils, or 2.54 mm, of aluminum**, a representative GEO figure. Over the five-year mission:

TID = 25 krad(Si); with the 2x rad-hard design margin, a 50 krad(Si) requirement

(10)

The factor-of-two margin for radiation-hardened parts is the standard guidance, with commercial parts carrying up to a factor of ten [14]. Selecting from the radiation hardness assurance categories [14], the requirement falls between category L at 50 krad(Si) and category R at 100 krad(Si), so **category R parts are specified**, a selection that holds even if the annual dose assumption is wrong by a factor of two. That insensitivity is what makes an assumed dose rate defensible at this stage of the design in place of a modeled one.

10.2 Mitigations

The parts baseline is category R at 100 krad(Si) for all avionics, with radiation-hardened CMOS preferred over the more susceptible technologies [14], backed by 100 mils of aluminum equivalent structural shielding and spot shielding on the few parts that cannot be procured in a hardened version. Over a 0.6 m avionics cube that shielding is 2.16 m² of surface and **14.8 kg of aluminum**, which is the shielding allocation inside the 75 kg C&DH line of Table 4. Against single-event effects the standard set applies: error detection and correction on all memory, watchdog timers, and redundancy with majority voting [14], with scrubbing cycles that refresh memory faster than upsets accumulate. Every power feed carries latch-up current limiting, so a latch-up trips a limiter and power-cycles the box instead of destroying it.

Underneath the hardware sits a formal radiation hardness assurance program, which identifies the exposure, sets margins, maintains a parts list, procures hardened parts where possible, and qualifies the remainder by test [14]. Two mission-level measures complete the picture. The SWPC feed from Section 5 gives warning of a proton event, so the vehicle safes non-essential avionics and holds proximity operations, and docking never begins inside a forecast SEP window. Separately, the array is sized so that post-degradation output still covers peak load, as Section 12.2 shows, rather than sized at beginning of life and allowed to fall short.

10.3 Impact on the other subsystems

As with plasma, the mitigations propagate outward, and **Table 11** accounts for them.

Table 11: Impact of the radiation mitigations on the rest of the vehicle.

Subsystem	Impact of the radiation mitigations
Structures and mass	14.8 kg of aluminum over the avionics volume plus spot shields is a direct mass charge, carried in Table 4.
Power	Degradation of 11.9% over five years drives the array from a beginning-of-life size to an end-of-life size, at 3,387 W BOL to hold 2,984 W at EOL (Section 12.2). Latch-up limiters add parts to every feed.
C&DH	Error correction, scrubbing, watchdogs, and voting logic cost throughput, memory, and software complexity. Category R parts are slower and far more expensive than commercial equivalents, which is why Table 8 carries \$12M against the avionics line.
ADACS	Star trackers are the most SEP-sensitive sensor on the vehicle, so the estimator carries inertial propagation through tracker dropouts (Section 11.3).
Thermal	Shielding mass adds thermal capacitance, which slightly damps the eclipse transient of Section 12.3. A minor benefit rather than a cost.
Operations	Servicing is suspended during SEP events, compounding with the plasma-driven docking constraint from Section 9.

11. Attitude Determination and Control (ADACS)

Sections 9 and 10 covered what the environment does to the vehicle's surfaces and electronics. It also pushes on the vehicle mechanically, and sizing the attitude control hardware means quantifying those pushes first.

11.1 The four disturbance torques

All four external disturbance torques were evaluated at GEO using the standard worst-case relations [3] with the Table 3 geometry. **Table 12** gives the formulas and results, and **Figure 8** shows how each varies with altitude.

Table 12: Worst-case external disturbance torques on MESA at GEO.

Disturbance	Formula [3]	Result (N-m)	Share
Solar radiation	$T_{sp} = F(c_{ps} - c_m), F = \frac{E_s}{c} A_s (1 + q) \cos i$	3.00×10^{-5}	93.5%

Disturbance	Formula [3]	Result (N·m)	Share
Gravity gradient	$T_g = \frac{3\mu}{2R^3} I_z - I_y \sin 2\theta$	1.53×10^{-6}	4.8%
Magnetic	$T_m = DB, B = M_E/R^3$ at the equator	5.31×10^{-7}	1.7%
Aerodynamic	$T_a = \frac{1}{2}\rho C_D A v^2 (c_{pa} - c_m)$	4.30×10^{-8}	0.1%
Worst-case total T_D		3.21×10^{-5}	100%

In these four relations F_s is the solar constant ($1,361 \text{ W/m}^2$), c the speed of light ($2.998 \times 10^8 \text{ m/s}$), A_s the illuminated area (m^2), q the surface reflectance factor, i the solar incidence angle, c_{ps} , c_{pa} , and c_m the centers of solar pressure, aerodynamic pressure, and mass (m), μ the Earth gravitational parameter (m^3/s^2), R the orbit radius (m), I_z and I_y principal moments of inertia ($\text{kg}\cdot\text{m}^2$), θ the off-nadir angle, D the residual magnetic dipole ($\text{A}\cdot\text{m}^2$), B the local field strength (T), M_E Earth's magnetic moment ($7.96 \times 10^{15} \text{ T}\cdot\text{m}^3$), ρ the neutral density (kg/m^3), C_D the drag coefficient, A the ram area (m^2), and v the orbital velocity (m/s).

Inputs: $A_s = 16.54 \text{ m}^2$, $q = 0.6$, $i = 0^\circ$, $c_{ps} - c_m = 0.25 \text{ m}$, $\theta = 10^\circ$, $|I_z - I_y| = 559 \text{ kg}\cdot\text{m}^2$, $D = 5 \text{ A}\cdot\text{m}^2$, $B = 1.06 \times 10^{-7} \text{ T}$, $\rho = 10^{-15} \text{ kg/m}^3$, and $v = 3,075 \text{ m/s}$. The SRP relation is normally written with a solar constant of $1,367 \text{ W/m}^2$; I use the $1,361 \text{ W/m}^2$ value from Section 4.1 everywhere for internal consistency, and the difference of four tenths of a percent does not move any conclusion.

$$T_D = 3.21 \times 10^{-5} \text{ N} \cdot \text{m}, \text{ of which SRP is } 93.5\%$$

(11)

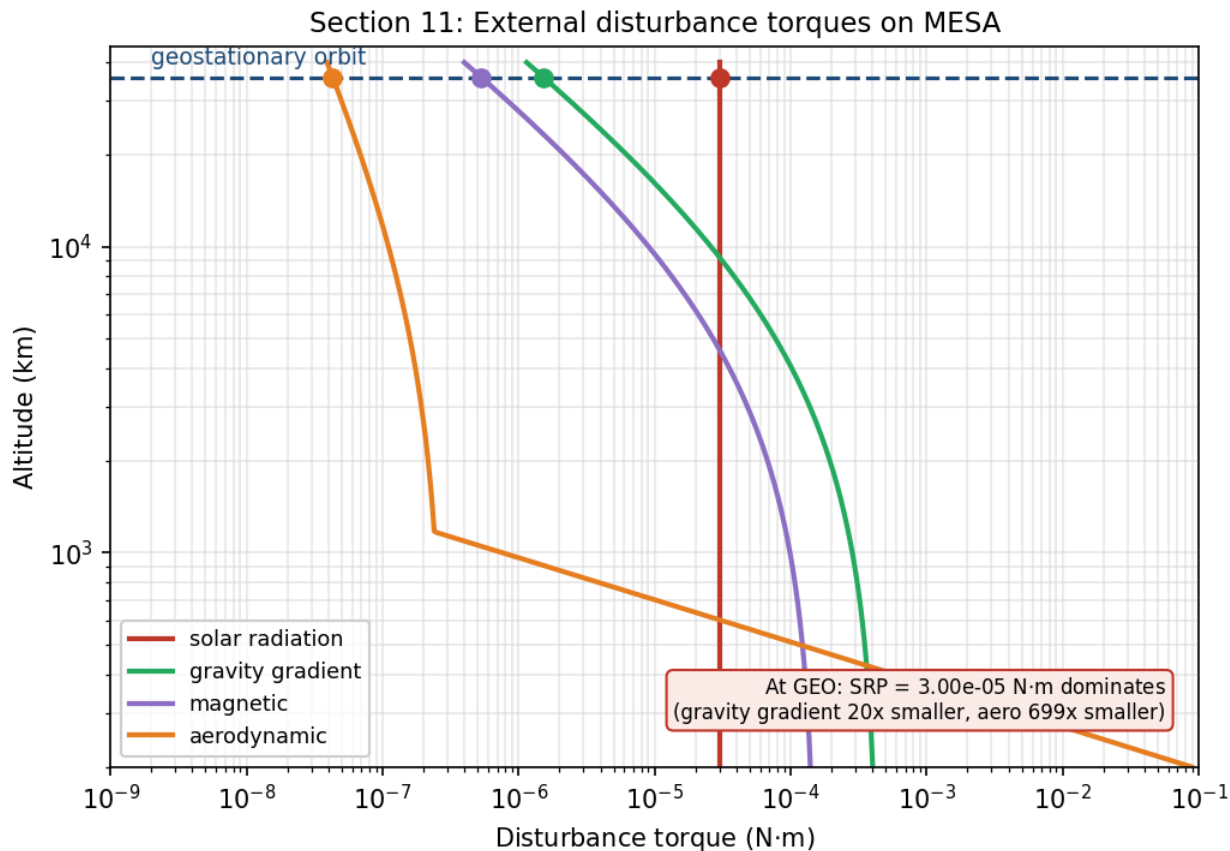


Figure 8: Torque magnitudes for the MESA geometry. Markers are the GEO operating point; SRP sets the ADACS design.

This is the expected GEO result, and it falls in the 10^{-6} to $10^{-4} \text{ N}\cdot\text{m}$ band that is typical for spacecraft disturbance torques [3]. Three points are worth drawing out. SRP dominates because it is the only one of the four that does not fall off with distance from Earth, since gravity gradient and magnetic torque both scale as R^{-3} and drag is gone entirely. Gravity gradient is modest here only because MESA is a compact bus with a small difference between I_z and I_y ; a long-boom vehicle at the same altitude would see far more. And aerodynamic torque at GEO is roughly seven hundred times smaller than SRP, which quantifies from a second direction the Section 3.3 conclusion that this is not a drag-driven design.

11.2 Actuator sizing

Disturbance rejection is not what sizes the wheels. Slewing the mated stack is, and **Table 13** compares every candidate driver on the same basis.

Table 13: Wheel sizing drivers [3]. The 30 degree slew in 300 s free-flying and 600 s mated is a derived requirement for reorientation during proximity operations and client inspection, not a customer-levied one; Appendix B gives its basis and sensitivity.

Driver	Relation	Result
Disturbance rejection	T_D	$3.21 \times 10^{-5} \text{ N}\cdot\text{m}$
30° slew, free flyer, 300 s	$M = 4I\theta/t^2$	$0.076 \text{ N}\cdot\text{m}, H = 11.4 \text{ N}\cdot\text{m}\cdot\text{s}$
30° slew, mated, 600 s	$M = 4I\theta/t^2$	$0.101 \text{ N}\cdot\text{m}, H = 30.3 \text{ N}\cdot\text{m}\cdot\text{s}$
Cyclic momentum storage	$H = 0.707 T_D (P_{orb}/4)$	$0.49 \text{ N}\cdot\text{m}\cdot\text{s}$
Secular momentum accumulation	$H = T_D P_{orb}$	$2.77 \text{ N}\cdot\text{m}\cdot\text{s per day}$

In these relations M is the control torque the wheels must produce (N·m), I the moment of inertia about the slew axis (kg·m²), θ the slew angle (rad), t the time allowed for the slew (s), H the angular momentum the wheels must store (N·m·s), T_D the worst-case disturbance torque from Eq. (11) (N·m), and P_{orb} the orbital period (86,164 s). The $4I\theta/t^2$ form assumes the standard accelerate-then-decelerate slew, half the time in each direction.

The mated slew at 0.101 N·m is the driving case, larger than the free-flyer slew and about three thousand times the disturbance torque, so the disturbance term does not participate in the sizing at all. Applying a one hundred percent margin factor to that driving case gives the wheel torque requirement in Eq. (12):

$$M_{RW} = M_{slew, mated}(1 + \text{margin}) = 0.101 \times 2 = 0.20 \text{ N} \cdot \text{m}$$

(12)

Justification for the one hundred percent margin. A factor of two on actuator torque is the standard allowance at this stage of design [3], and on this vehicle three specific unknowns justify carrying it rather than trimming it.

The first is the client. The 3,000 kg client in Table 3 is an assumption, and the mated inertia that sizes the wheels scales with whatever actually gets captured. Re-running the sizing across client masses shows what the margin buys: **0.20 N·m covers a client of up to about 14,100 kg** at the same 600 s slew, which is well beyond the heaviest satellite in the GEO belt. Without the margin the requirement is met only by the assumed client and nothing above it.

The second is that capture is not a nominal maneuver. A client with degraded attitude control arrives with residual body rates, and the combined center of mass is not known precisely until the two vehicles are mated and the stack is characterized on orbit. Both effects raise the torque actually demanded above the value a clean slew calculation predicts.

The third is schedule authority. Held at 0.20 N·m, the baseline 3,000 kg client can be slewed in **425 s rather than 600 s**, a thirty percent faster maneuver. That matters during proximity operations, where the ability to reorient quickly is what converts an off-nominal approach into an abort instead of a collision.

The margin is therefore not a round number applied out of habit. It is what makes the design insensitive to the one parameter, client mass, that the customer rather than the contractor controls.

At above 0.15 N·m this lands in the large-satellite class, which carries **25 kg and 100 W per wheel** [17]. The baseline is **four wheels in a pyramid**, which gives one-fault tolerance and three-axis control from any three of them.

The momentum behavior is the more interesting result. Because SRP at GEO is nearly constant in inertial space rather than cycling with orbit position, the stored momentum is **secular**: it ramps at 2.77 N·m·s per day instead of averaging out over an orbit. With a 50 N·m·s dump threshold against a 200 N·m·s wheel capacity, that is a dump roughly **every eighteen days**, or 101 dumps across the mission, as shown in **Figure 9**.

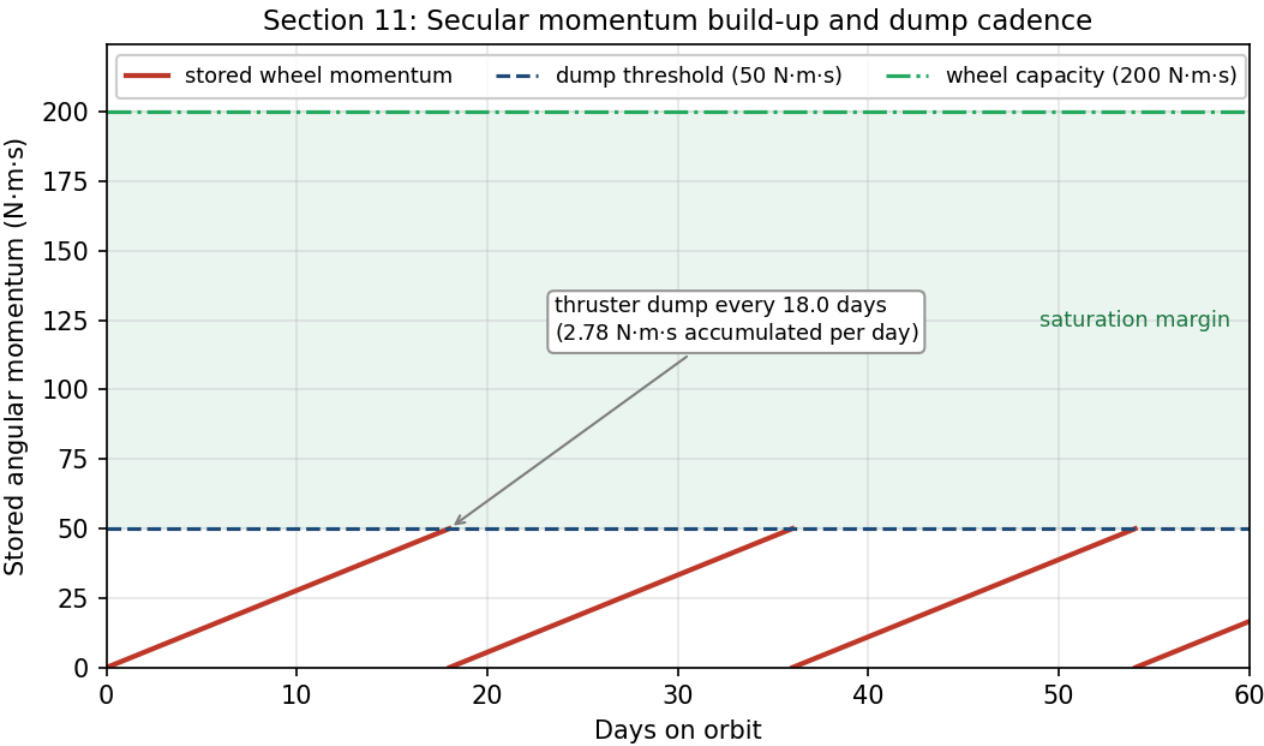


Figure 9: SRP torque at GEO is secular, so the wheels ramp and must be desaturated with thrusters, not magnetorquers.

Momentum dumping cannot use magnetorquers. Earth's field at GEO is only 1.06×10^{-7} T, so unloading 50 N·m·s in any reasonable time would demand an impractically large dipole, and magnetic torquers are simply not useful in high-Earth orbit [3]. **Dumping is done with the hydrazine thrusters**, and Table 7 charges it 5.2 kg of propellant across the mission rather than treating it as free.

11.3 Sensors, mass, and power

Table 14 lists the subsystem as flown.

Table 14: ADACS mass and power. Actuator and sensor values are the GEO-class figures [3], [17]; the RPO sensor line is an engineering estimate carried from the Section 2.3 requirement.

Component	Qty	Mass (kg)	Power (W)
Reaction wheels, 200 N·m·s class	4	100.0	120 avg / 400 peak
Star trackers	2	10.0	36
Inertial measurement units (one active)	2	10.0	30
Coarse sun sensors	4	4.0	12
ADACS control electronics	1	8.0	25
RPO LIDAR and cameras	1 set	25.0	60
Total		157.0	283 avg

ADACS: 157 kg (7.8% of wet mass), 283 W orbit-average, 563 W peak during a mated slew

(13)

Two sensor choices follow directly from the environment sections. Star trackers are doubled because they are the most SEP-sensitive sensor on the vehicle, as Section 10.1 established, and the estimator propagates on the inertial measurement unit through tracker dropouts rather than losing attitude knowledge mid-approach. GPS is not baselined, because at GEO the vehicle sits above the constellation and would depend on sidelobe tracking, so ground-based ranging is the primary orbit determination source with GPS sidelobe reception held as a possible later upgrade.

12. Simulations: Orbit, Power, and Thermal

The preceding sections quoted results from a common analysis. Everything in this report comes from one script, `spce5065_final_figs.py`, which computes the mass properties, the budgets, the thermal balance and transient, the disturbance torques, the wheel sizing, the power profile, and the meteoroid flux, and writes every figure except the three STK captures. This section presents the simulations themselves: the orbit, the power profile over a full day, the thermal transient through eclipse, and an independent rebuild of the mission in Systems Tool Kit.

12.1 Orbit simulation

Figure 10 simulates the MESA orbit, propagated numerically over one sidereal day, which is the GEO period. Panel A is a three-dimensional inertial rendering: Earth is drawn to scale with its spin axis and equator marked, the geostationary and graveyard orbits are shown as the equatorial rings they are, and the satellite is stepped hourly to show one revolution per sidereal day, with MESA and a client separated in longitude along the belt. Panel B is the resulting ground track.

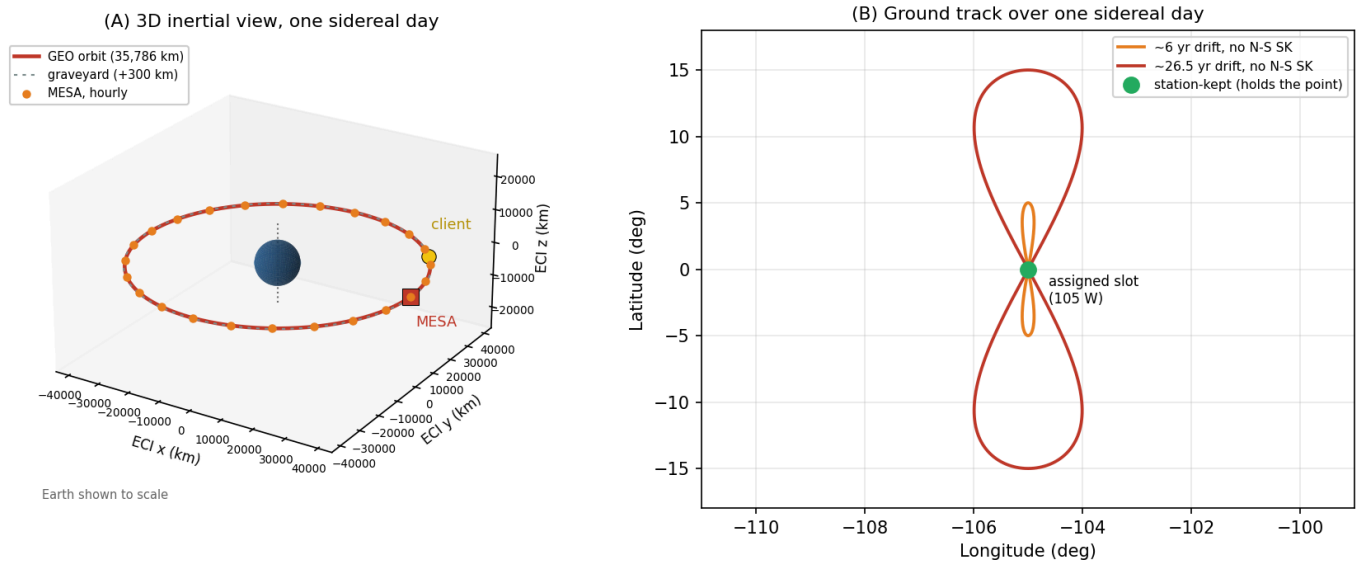


Figure 10: Three-dimensional simulation of the MESA orbit over one sidereal day. The vehicle holds one longitude, but without north-south stationkeeping the inclination grows and the ground track opens into an analemma.

A stationkept satellite holds a single point over its assigned longitude, taken here as 105° W, and that is the green marker in Panel B. Without north-south stationkeeping, luni-solar gravity grows the inclination at the rate quantified in Section 3.3 and the ground track opens into the classic figure-eight analemma, reaching roughly $\pm 5^\circ$ of latitude after a few years and $\pm 15^\circ$ after about 26.5 years. That drift is precisely why the stationkeeping allocation in Table 6 is the largest single line in the delta-v budget: MESA is not fighting decay, it is fighting drift, for itself and for every client it is holding.

12.2 Power, eclipse, and battery sizing

Figure 11 plots array output over one day at equinox, which is the worst case of the year for both eclipse duration and battery depth of discharge.

Section 12: MESA array power over one day at GEO (equinox)

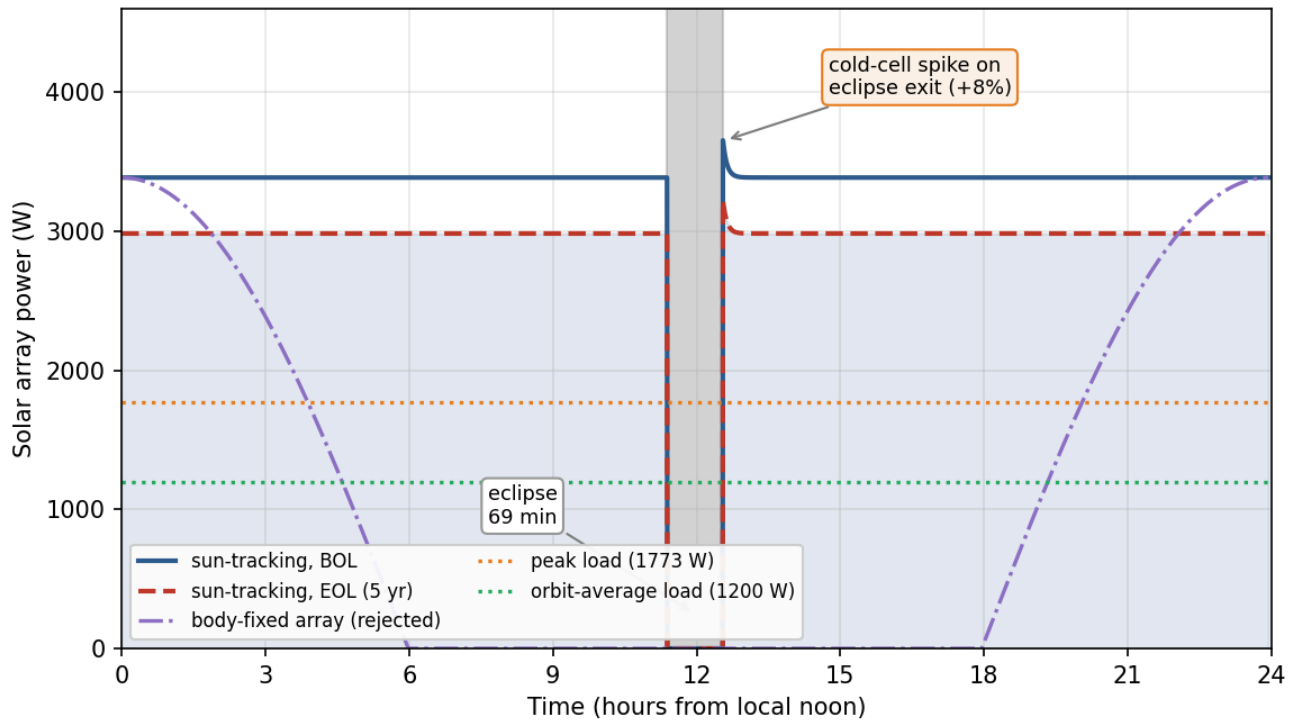


Figure 11: Single-axis tracking holds full output through the day; a body-fixed array would not, which is why the wings track.

The eclipse is the deepest of the year, and from cylindrical shadow geometry the satellite spends

$$t_{\text{eclipse}} = 69.4 \text{ min in shadow at equinox}$$

(14)

which is close to the commonly quoted maximum of about 72 min, the difference being the penumbra and the Sun's finite disk, neither of which the cylindrical model carries. Section 12.4 confirms both figures against STK.

Battery capacity follows directly from Eq. (14). Carrying the full 1,773 W peak load through 69.4 min is 2,051 W·h of delivered energy, and at a 60 percent depth of discharge with a 90 percent discharge-path efficiency the installed capacity must be

$$E_{\text{batt}} = \frac{2,051}{0.60 \times 0.90} = 3,798 \text{ W} \cdot \text{h} \rightarrow 38 \text{ kg installed at } 150 \text{ W} \cdot \text{h/kg cells}$$

(15)

where E_{batt} is the installed capacity (W·h), 2,051 W·h is the energy delivered to the peak load across the Eq. (14) eclipse, 0.60 is the allowed depth of discharge, and 0.90 is the efficiency of the discharge path from cell to load,

using a 1.5 packaging factor from cells to installed battery [3]. That 38 kg is the battery line inside the 268 kg power allocation of Table 4. Sizing to 60 percent depth of discharge rather than deeper is a cycle-life decision. Earth subtends a half-angle of 8.70° from GEO, so eclipses occur only while the solar declination sits inside that, which is about 44 days either side of each equinox and gives roughly eighty-eight eclipses a year. The battery therefore sees on the order of four hundred and forty deep cycles across five years, and depth of discharge is the strongest lever on how many of those it survives.

Three features of the profile are worth noting. First, the arrays use single-axis sun tracking, which is why the output stays flat rather than following the cosine curve of the body-fixed case also plotted; that comparison is the justification for accepting the mass and mechanism complexity of tracking wings. Second, coming out of eclipse the cells are cold and briefly produce about eight percent above nominal before warming to steady state, which the power regulation must accept without tripping. Third, and this is the budget result:

$$P_{\text{BOL}} = 3,387 \text{ W} \rightarrow P_{\text{EOL}} = 2,984 \text{ W after five years (88.1\%)}$$

(16)

At 2.5 percent per year the array loses 11.9 percent over the mission, leaving 68 percent margin over the 1,773 W peak load at end of life. Extrapolating that decay, array output does not fall to the peak load until roughly 25.6 years, so power is emphatically not the life-limiting mechanism, and neither is drag.

12.3 Thermal transient through eclipse

Eq. (7) gives two steady-state temperatures, but a 69.4 min eclipse is short compared with the thermal time constant of a 1,400 kg vehicle, so the bus never reaches the cold bound. **Figure 12** integrates the lumped-capacitance form of Eq. (6), $C dT/dt = Q_{\text{in}}(t) - \sigma \epsilon AT^4$, through one eclipse for two very different thermal masses.

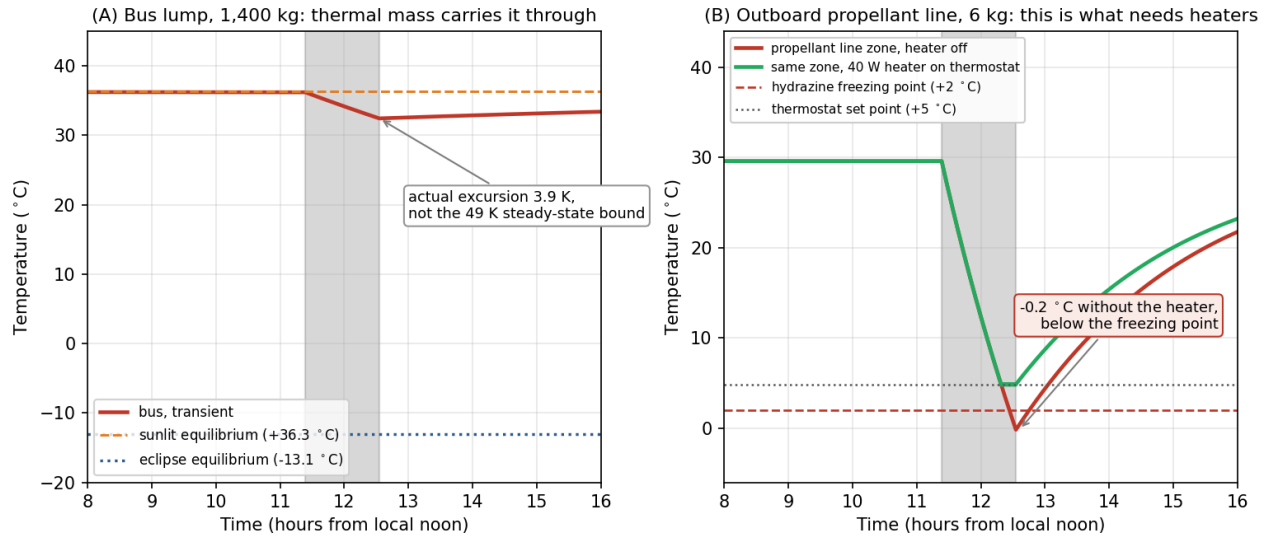


Figure 12: Transient simulation through the equinox eclipse. The bus barely moves; the low-mass outboard zones set the heater requirement.

Taking the bus as an isothermal lump of the 1,400 kg dry mass at an aluminum-dominated 900 J/(kg·K) gives $C = 1.26 \times 10^6$ J/K against a radiative conductance of 31.1 W/K at the sunlit temperature, so the time constant is **11.3 hours against a 1.16 hour eclipse**. Panel A is the consequence: the bus drops **3.9 K**, from +36.3 °C to +32.4 °C, and recovers within a couple of hours of sunrise. It never approaches the -13.1 °C steady-state bound.

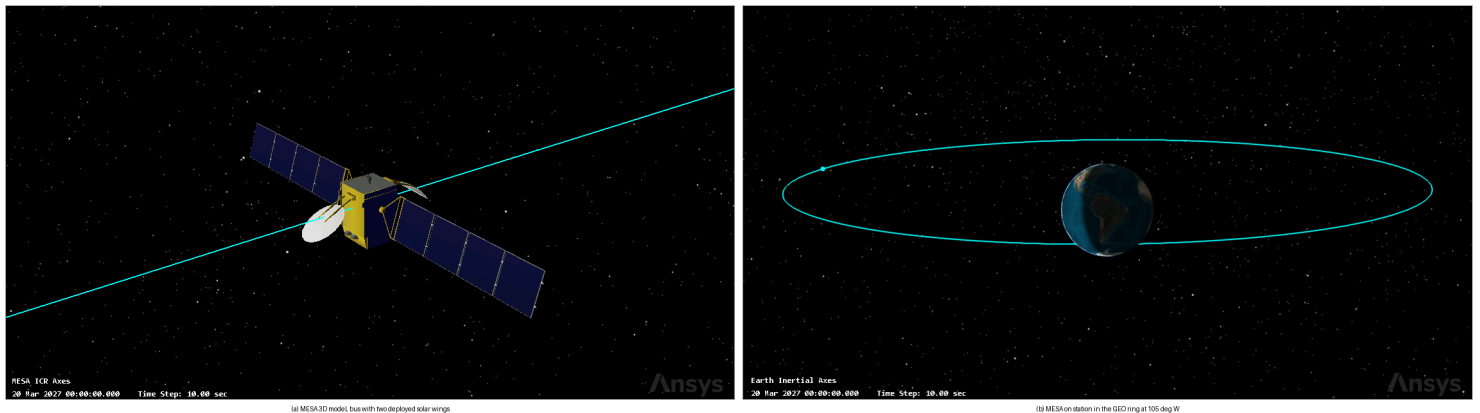
Panel B is the zone that actually needs the heaters. An outboard propellant line and latch-valve assembly, taken as 6 kg with an effective εA of 0.10 m² and no internal dissipation, has a time constant of 2.4 hours, comparable to the eclipse itself. It sits at +29.6 °C in sunlight and falls to **-0.2 °C** by the end of eclipse with the heater off, which is below hydrazine's 2 °C freezing point. Holding it at the +5 °C set point needs 33.9 W of steady power, so the 40 W heater string on that zone closes the case with margin and the simulated minimum becomes +4.8 °C.

This is the single most useful result in the simulation set, because it changes where the hardware goes rather than just confirming a number. The 260 W heater allocation is real, but it belongs on the battery, the propellant lines and tanks, the docking mechanism, and the arm joints, not on the bus. Two caveats keep it honest: a perfectly isothermal 1,400 kg lump is optimistic, since real vehicles have gradients and imperfectly coupled boxes, and the outboard zone's εA is an estimate. Both are exactly the assumptions the thermal-balance test in Section 8.4 exists to correlate.

12.4 Independent verification in STK

Everything above comes from one Python model, so it is internally consistent by construction but not independently confirmed. The mission was therefore rebuilt in Systems Tool Kit 13.1.0 [18] and the eclipse, the array output, and the orbital lifetime were re-derived there. The scenario runs one day from 20 March 2027, the vernal equinox, which is the worst-case eclipse season and the case Eq. (14) describes.

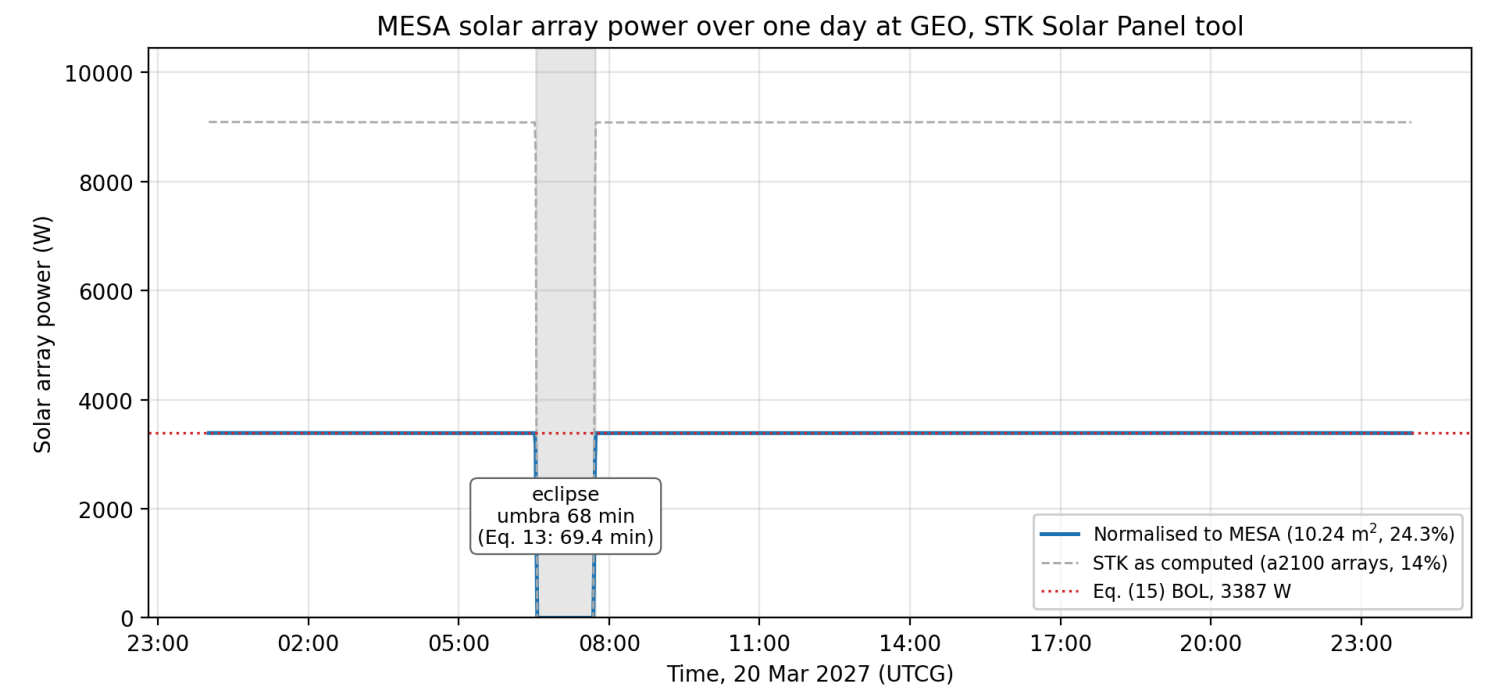
The vehicle is a two-body propagation at the Eq. (1) semi-major axis, circular and equatorial, holding the 105° W slot. Sampling the subsatellite point across the day confirms it is geostationary rather than merely the right size: longitude holds at 105.0000° W with a spread of 0.0005° over twenty-four hours. **Figure 13** shows the scenario, with the vehicle model at left and the full GEO ring at right. The model is a geostationary communications bus with two deployed wings, the closest available match to the 1.8 by 1.8 by 3.5 m MESA bus carrying 10.24 m² of array.



The Solar Panel tool computes illuminated area by rendering the vehicle against the Sun and integrating the lit surface, then applies

$$P = \eta \cdot S \cdot A_{eff} \cdot G_s$$

(17)



The eclipse is the meaningful cross-check, since it depends only on the orbit and the epoch and not on the model. STK gives **68.0 min** of full umbra and **72.0 min** of total shadow including penumbra, centered on 07:07 UTC, which is local midnight at 105° W. The 69.4 min cylindrical estimate of Eq. (14) falls between the two, exactly where it should: that model has no penumbra and treats the Sun as a point, so it overstates umbra and understates total shadow. The 72.0 min figure also confirms the commonly quoted maximum that Section 12.2 asserted without a source.

The power comparison is weaker, and no more is claimed from it than it supports. STK takes array area and efficiency from the model geometry rather than from Table 3, so its raw 9,085.7 W describes that bus, not MESA. Inverting Eq. (17) gives an implied area of 47.68 m², and rescaling to MESA's 10.24 m² at the Section 12.2 efficiency of 24.3 percent returns 3,386.9 W. That matches Eq. (16) to within a watt, but both sides evaluate the same product of efficiency, area, and irradiance, so the agreement is close to definitional. What it does establish is that STK uses the same solar constant and formulation, and that Eq. (16) is arithmetically sound.

The Lifetime tool closes out Section 3.3. Given the Table 3 drag area of 16.54 m², the drag coefficient of 2.2, and the 2,000 kg wet mass, STK reports no decay within a 36,500 day limit, or one hundred years, which is the result reproduced in **Figure 15** alongside the inputs that produced it.

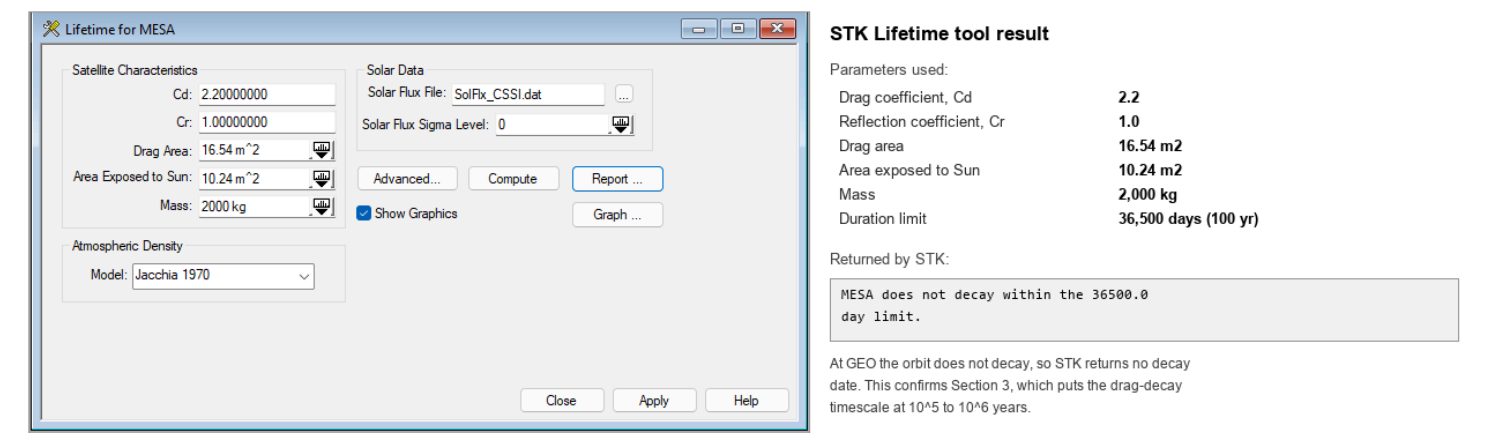


Figure 15. STK Lifetime tool: MESA input parameters (left) and the returned result (right).

That is the correct answer at GEO, not a tool failure: a decay timescale of 10^5 to 10^6 years is indistinguishable from never for any run bounded at a century.

Table 15: Cross-checks between the Python model and STK.

Quantity	This report	STK 13.1.0	Assessment
Eclipse, umbra at equinox	69.4 min, Eq. (14)	68.0 min	independent, agrees within two percent

Quantity	This report	STK 13.1.0	Assessment
Eclipse, umbra plus penumbra	about 72 min, quoted	72.0 min	independent, confirms the quoted maximum
Array output at BOL	3,387 W, Eq. (16)	3,386.9 W, normalized	consistent, but not independent
Orbital decay at GEO	none, 10^5 to 10^6 yr, Eq. (3)	does not decay in 100 yr	independent, confirms Section 3.3

Two of the four are independent confirmations, one is a consistency check labeled as such, and none contradict the analysis, as **Table 15** sets out. The scenario ships alongside this report as `MESA_MS2.zip`.

13. Integrated Risk Assessment and Mission Assurance

Sections 8 through 12 each closed their own subsystem. A customer buying this vehicle is buying the combination, so this section assembles the risk picture: the one environmental threat the earlier sections did not quantify, the failure modes around the docking sequence, and where the whole design sits before and after its mitigations.

13.1 Micrometeoroids and orbital debris

GEO is the benign regime for man-made debris and the ordinary one for meteoroids. Catalogued objects in the belt drift slowly relative to a stationkept satellite, on the order of metres per second, so a conjunction there is a collision avoidance problem rather than a hypervelocity one, and the population density is orders of magnitude below LEO. Sporadic meteoroids are the opposite: they arrive at roughly 20 km/s regardless of altitude, and the flux at GEO is essentially the interplanetary flux.

Using the Grun sporadic flux [19] with the Earth-shielding and gravitational-focusing factors, $\chi = (1 + \cos \theta)/2$ for randomly oriented surfaces and $G = 1 + R_a/r$ with $\sin \theta = R_a/r$ and $R_a = R_E + 100$ km, GEO gives $\chi = 0.9941$ and $G = 1.1536$, so $\chi G = 1.147$. The same vehicle at 400 km gives $\chi = 0.6471$ and $G = 1.9557$, so $\chi G = 1.266$. That is the counterintuitive and useful result: Earth shielding and gravitational focusing very nearly cancel, so the **net meteoroid flux is within about ten percent of orbit-independent**, which is the exact opposite of how man-made debris behaves. Moving MESA to GEO buys enormous relief from debris and almost none from meteoroids.

Taking MESA's exposed area as 52.2 m², the 31.68 m² bus exterior plus both faces of both wings, and applying the Poisson relation $p = 1 - e^{-FAt}$, in which p is the probability of at least one impact, F the cumulative flux of particles at or above the stated mass (m⁻²yr⁻¹), A the exposed area (m²), and t the exposure time (yr), over five years gives **Table 16** and **Figure 16**.

Table 16: Meteoroid flux and five-year impact probability at GEO for 52.2 m² of exposed area.

Particle diameter	Mass (g)	Flux (m ⁻² yr ⁻¹)	Mean interval	$P(\text{at least one in 5 yr})$
0.1 mm	1.31×10^{-6}	1.33	5.3 days	~100%
1 mm	1.31×10^{-3}	4.86×10^{-4}	39 yr	11.9%
5 mm	1.64×10^{-1}	8.91×10^{-7}	2.2×10^4 yr	0.023%
10 mm	1.31	5.63×10^{-8}	3.4×10^5 yr	0.0015%

$$P_{5\text{yr}}(d \geq 1\text{ mm}) = 11.9\%, \quad P_{5\text{yr}}(d \geq 5\text{ mm}) = 0.023\%$$

(18)

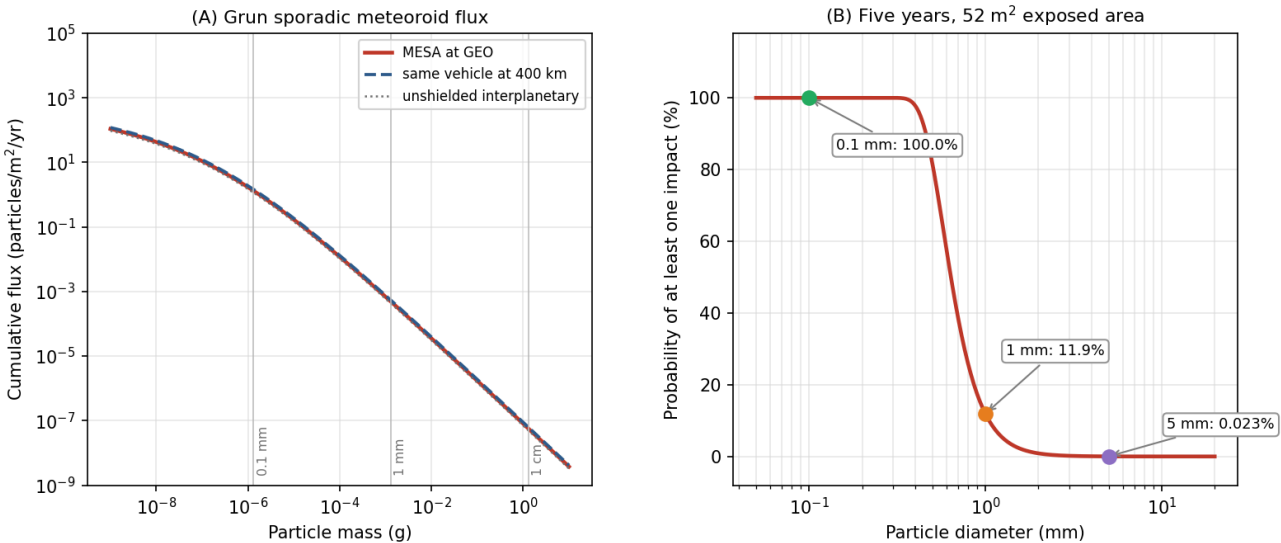


Figure 16: Sub-millimetre impacts are certain and are designed around; the mission-ending sizes are rare enough to accept.

The design response follows the shape of that distribution rather than trying to armor against all of it.

- **Sub-millimetre impacts are certain and are absorbed by design.** Thousands of 0.1 mm hits over five years pit the coatings, degrade the optical properties of the OSR radiator, and cost a small amount of array output. This is carried as part of the 2.5 percent per year array degradation and in the coating end-of-life properties used in Section 8, not as a discrete risk.
- **Millimetre impacts are likely and are handled by redundancy rather than shielding.** A 1 mm particle will very probably hit somewhere in five years. Array string isolation diodes, already required for arc tolerance in Section 9.2, mean a punctured cell string is lost rather than the wing.
- **The pressurized items get a bumper.** For the propellant tanks, where a penetration is not a degradation but a loss of vehicle, a Whipple shield was sized [14] against a 0.3 cm meteoroid at 20 km/s with a 10 cm standoff and an Al 6061-T6 rear wall at 35 ksi: bumper thickness $t_b = 0.56$ mm and rear wall $t_w = 3.6$ mm. That is a modest, entirely conventional shield, and the tank walls already provide much of it.
- **The RPO optics get covers.** The docking cameras and LIDAR are the mission, and they are only exposed during proximity operations, which is a few percent of the mission. Closing covers outside those windows removes most of their integrated exposure to both meteoroids and contamination for almost no mass.

13.2 Failure modes around the docking sequence

The docking sequence concentrates the mission's risk into a few minutes. Table 17 lists each failure mode against the section that owns its mitigation, so only the three design rules that exist because of docking, and nowhere else in the report, are set out here.

Abort criteria are set at the mechanism's qualified envelope, not at the controller's optimistic one. A client that has lost attitude control may be tumbling faster than the capture mechanism can accept, and its rates are not known until they are measured. The far-field inspection phase in Figure 1 exists to measure them before anything is committed.

A loss of attitude knowledge must produce a drift-away, not a closing trajectory. A solar particle event takes out the star trackers at the worst possible moment, so the approach geometry is designed to be passively safe: if the estimator loses the client, the relative motion opens rather than closes.

The release path is redundant. A capture mechanism that will not let go strands MESA on its first client and converts a reusable asset into a single-use one, which is a worse outcome than failing to capture at all.

13.3 Risk register and residual posture

Table 17 collects the docking failure modes above, together with the environmental and wear-out risks from the rest of the report, into a register with the likelihood and consequence scoring used in **Figure 17**, where each risk is plotted before and after its mitigation.

Table 17: MESA risk register. Likelihood and consequence run 1 (low) to 5 (high).

ID	Risk	Before	Mitigation	After	Owner section
R1	ESD at capture damages both vehicles	4 x 5	Plasma contactor, equalize before contact, bonded interface	2 x 3	9.2
R2	SEP event during proximity operations	3 x 4	SWPC gating, IMU propagation, passively safe aborts	2 x 2	5, 10.2
R3	Deep-dielectric arc into the harness	3 x 4	100 mil shielding, conductive dielectrics, EMI filtering	2 x 2	9.2
R4	Client tumbling beyond the capture envelope	3 x 5	Far-field inspection, rate-based abort criteria	2 x 3	13.2
R5	RPO optics contaminated or hazed	4 x 4	E595 screening, bakeout, TVAC, optic covers, contactor	2 x 2	6, 13.1
R6	MMOD strike on a solar wing	3 x 2	String isolation, degradation carried in the EOL budget	3 x 1	13.1
R7	Cold welding in the capture mechanism	2 x 5	Vacuum qualification, dry lubricants, redundant release	1 x 3	6, 8.4
R8	Wheel saturation from secular SRP torque	4 x 2	Four-wheel pyramid, thruster dumps every 18 days	1 x 2	11.2
R9	TID above the 5 krad(Si)/yr assumption	2 x 4	Category R parts at 100 krad(Si), a 4x margin on the estimate	1 x 2	10.2
R10	Heater string failure freezes a propellant line	2 x 4	Redundant thermostats, cross-strapped strings, zoning	1 x 2	8.2, 12.3

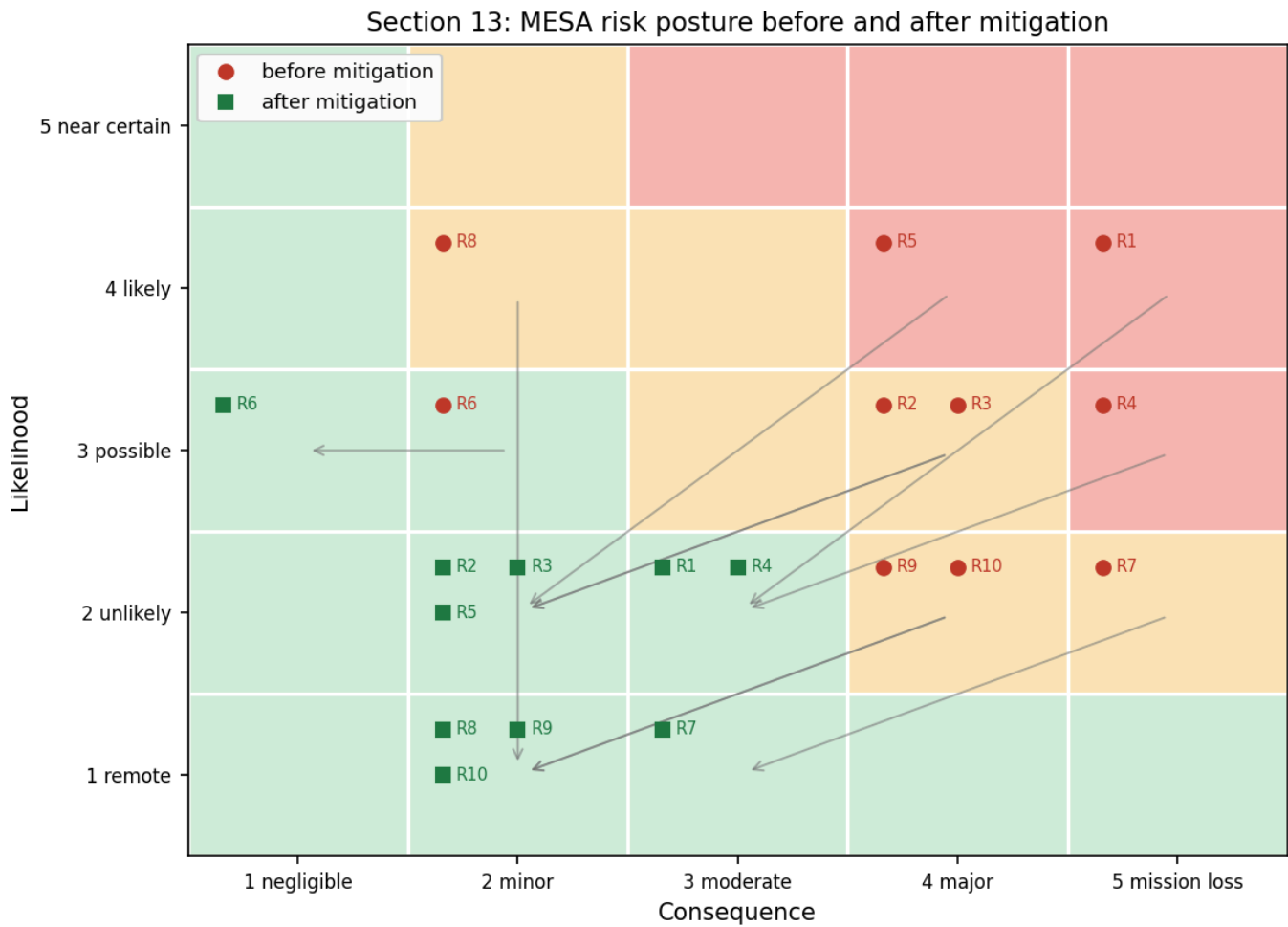


Figure 17: Every mitigation in this report moves a specific risk down and to the left. Nothing remains in the red band.

Two observations matter more than the individual scores. First, every mitigation in this report moves a specific risk down and to the left, and after mitigation nothing remains in the red band, with the highest residual scores belonging to R1 and R4, both of which are docking risks and both of which are managed operationally as well as in hardware. Second, R6 is the one risk whose likelihood does not move: a meteoroid will hit the arrays, and the design response is to make the consequence negligible rather than to pretend the event is avoidable. That distinction between reducing likelihood and reducing consequence is the honest way to present a risk posture.

13.4 The verification program this implies

Pulling the sections together, the design depends on four things being verified on the ground rather than discovered on orbit: the thermal-balance correlation of Section 8.4, the vacuum qualification of the capture mechanism and deployables from Section 6, the grounding-continuity and EMI verification implied by Section 9.2, and the radiation hardness assurance test program of Section 10.2. All four sit inside the \$9M integration and test line of Table 8, and all four exist because MESA gets exactly one chance at each client.

14. Conclusions

MESA is a 2,000 kg geostationary servicing tug, and this report has carried it from the environment it must survive through to a design that closes against that environment with margin in every budget.

The orbit is geostationary because the clients are there, and the environmental trade that decision buys is favorable. GEO removes atmospheric drag, atomic oxygen, and the dense debris population, at the cost of worse spacecraft charging and worse energetic-particle radiation. Drag-decay timescales at GEO run to 10⁵ to 10⁶ years, five orders of magnitude beyond the requirement, and STK independently confirms no decay within a century. The vehicle is not fighting decay, it is fighting drift, at about 50 m/s per year.

The thermal control system is passive-dominant: MLI over the bus, 4.9 m² of OSR radiator on the anti-sun face, zoned heaters, and heat pipes to the dissipating boxes. The bus equilibrates at +36.3 °C in sun and -13.1 °C in eclipse, a steady-state swing of 49.4 K, and the transient simulation shows the bus itself moves only 3.9 K through a

real eclipse while an outboard propellant line falls to -0.2 °C, below hydrazine's freezing point. That result is why the 260 W of heater power is zoned onto low-mass outboard hardware rather than spread across the bus.

The plasma environment floats the vehicle near -2.16 kV during substorms, where the real threat is differential charging into ESD rather than the absolute potential. The mitigations are conventional, comprising conductive bonded surfaces, ITO coatings, and single-point grounding, with one exception specific to this mission: because docking bonds two independently charged vehicles, MESA carries a plasma contactor and equalizes potential before mechanical contact rather than through the capture latch.

The radiation environment delivers roughly 25 krad(Si) over five years behind 100 mils of aluminum, which with the standard factor-of-two margin sets a 50 krad(Si) requirement and drives category R part selection at 100 krad(Si). That selection is insensitive to a factor-of-two error in the dose assumption, which is what makes it defensible at this stage. The array loses 11.9 percent to displacement damage, and that loss is what sizes the array at beginning of life.

The ADACS is set by solar radiation pressure, which contributes 93.5 percent of the 3.21×10^{-5} N·m worst-case disturbance torque, but the wheels are actually sized by slewing the mated stack, since capturing a 3,000 kg client raises I_x by a factor of 5.3 and drives a 0.20 N·m wheel requirement, giving four wheels inside a 157 kg, 283 W subsystem. Because SRP at GEO is secular rather than cyclic, momentum ramps at 2.77 N·m·s per day and must be dumped with thrusters roughly every eighteen days, since Earth's field at GEO is too weak for magnetorquers to be useful.

Every budget closes. Mass closes at 2,000 kg with 150 kg of margin. Power closes with 68 percent margin over peak load at end of life and a 38 kg battery sized on a 69.4 min equinox eclipse. Propellant closes at 165 kg consumed of 600 kg loaded, with the hydrazine tank as the binding sub-budget at about ten client servicing cycles against a baseline manifest of six. Cost closes at the \$100M cap with \$5M in reserve. One refinement to Milestone 2 belongs here: with electric propulsion at 1,600 s, propellant sizes the **servicing capacity** rather than the calendar life, and what bounds a five-year contract is wear-out of the wheels, the docking mechanism, and the battery. That shifts the risk from tankage to qualification testing, which is the better place for it.

The unifying result is that docking, not free flight, drives this design. The mated inertia sizes the wheels. The docking interface creates a charging failure mode that no conventional satellite has, and it is the highest residual risk on the register even after mitigation. The RPO sensors are what make contamination control and vacuum testing non-negotiable. The vehicle spends eighty percent of its life attached to something else, and every subsystem shows it.

That is also the argument for awarding this contract. A GEO servicer is not a communications satellite with an arm attached, and a design that treats it as one will size its wheels for the wrong inertia, ground its exterior for the wrong failure mode, and put its heaters in the wrong place. MESA is sized for the mission it actually flies, its budgets close with margin, its analysis is reproducible from a single script, and its central results have been checked against an independent tool.

Appendix A: Changes since Milestone 2

Corrections to Milestone 2, which scored ninety-two out of one hundred. Every deduction is addressed in the body of the report rather than listed separately.

Deduction	Correction	Where
Thermal control, add margin	A margin assessment now states the predicted temperature, the component limit, and the margin in each case, works the internal-load growth allowance, and shows why oversizing the radiator is a trade against heater power rather than a free improvement	Section 8.2, Table 9
Plasma, the floating potential was quoted rather than derived	The current-balance argument behind the 2.50 coefficient is now given, the electron temperature that goes into it is stated, and the arithmetic is shown and checked	Section 9.1, Eqs. (8) and (9)
ADACS, no justification for the one hundred percent margin factor	The margin is now defended in terms of what it buys: a client of up to 14,100 kg at the same slew time, tolerance to off-nominal capture, and a thirty percent faster slew on the baseline client	Section 11.2
Grammar, include units in the nomenclature	The nomenclature is split into acronyms and symbols, and every symbol carries its units	Nomenclature
Grammar, define all variables with equations	Every display equation and every formula given in a table is followed by a definition of its variables	throughout
Grammar, do not use first person in a formal report	The report is written impersonally throughout	throughout

New work added past Milestone 2.

Change	Where
Report reorganized into the final report structure, with the orbit decision presented before the environment sections that justify it	throughout
Mass, delta-v, propellant, and cost budgets added and closed, which Milestone 2 listed as remaining work	Section 7
Concept of operations added, so the design decisions tie to a specific mission timeline	Section 2.4

Change	Where
Transient thermal simulation added; it shows the bus moves 3.9 K through eclipse against the 49.4 K steady-state bound, which changes where the heaters go	Section 12.3
Micrometeoroid and debris assessment added, with a Whipple shield sized for the propellant tanks	Section 13.1
Docking-sequence failure modes collected into a risk register and a risk matrix	Sections 13.2 and 13.3
Milestone 2's conclusion that propellant is the life limiter refined: propellant sizes the servicing capacity, and the five-year calendar life is set by wear-out items	Sections 7.4 and 14
Table of contents rebuilt with page numbers and hyperlinks; numbers of ten or below spelled out in running text	throughout

Appendix B: Assumptions register

Design at this stage rests on values that are chosen rather than derived, and a reader is entitled to know which is which. Every assumption that materially affects a result in this report is listed below with its basis, what it drives, and what happens if it turns out to be wrong. Values that are computed rather than assumed, such as the moments of inertia or the equilibrium temperatures, are not listed; they follow from the rows below.

Assumption	Value	Basis	What it drives	If it is wrong
Bus envelope	1.8 x 1.8 x 3.5 m	Chosen to package tanks, arm, and avionics inside a launcher-compatible envelope	I_x, I_y, I_z ; 31.68 m ² radiating area; 6.30 m ² projected area	Scales the inertias and both areas, so thermal and torque results move together
Solar wing size and mass	2 x 1.6 x 3.2 m, 60 kg each	Sized to clear the end-of-life load with margin; mass is a class figure	Array output, illuminated area A_s, I_x	Array power and SRP torque are both linear in area
c_{ps} to c_m offset	0.25 m	Engineering estimate for a vehicle carrying an arm and docking hardware on one face	SRP torque, hence 93.5% of T_D , the momentum rate, the dump cadence, and the dump propellant	The most sensitive assumption in the report: SRP torque is linear in the offset
Slew requirement	30° in 300 s free-flying, 600 s mated	Derived requirement for reorientation during proximity operations and client inspection	Wheel torque of 0.20 N·m, which is the report's central sizing claim	M scales as $1/t^2$; a 425 s mated slew consumes the whole margin (Section 11.2)
Client mass	3,000 kg	Representative GEO communications satellite at end of life	Mated inertia, wheel sizing	Section 11.2 shows 0.20 N·m covers a client up to 14,100 kg
Client envelope and docking standoff	2.0 x 2.0 x 3.0 m, centre of mass 3.0 m from MESA's	Assumed for the mated inertia calculation	I_x mated = 17,329 kg·m ² , 1.80 m centre-of-mass shift	The standoff dominates through the parallel-axis term
Residual magnetic dipole D	5 A·m ²	Typical for this vehicle class	Magnetic torque, 1.7% of T_D	Negligible unless wrong by an order of magnitude
Reflectance factor q	0.6	Rough-estimate value [3]	SRP force and torque	Linear in $(1 + q)$; real surfaces range 0.2 to 0.9
Off-nadir angle θ	10°	Worst-case pointing excursion assumed for the gravity gradient bound	Gravity gradient torque, 4.8% of T_D	Enters as $\sin 2\theta$; small effect on the total
Wheel capacity and dump threshold	200 and 50 N·m·s	Class figures for the selected wheel size	Eighteen-day dump cadence, 5.2 kg of dump propellant	Cadence and propellant scale with the threshold
Thruster moment arm	0.9 m	Bus half-width	Momentum-dump propellant	Linear; 5.2 kg is small against the 80 kg hydrazine load
Cell efficiency chain	0.30 BOL x 0.90 packing x 0.90 hot derate	Triple-junction GaAs class figures	3,387 W at beginning of life	Output is linear in the product
Array degradation	2.5% per year	Standard GEO allowance	2,984 W EOL, an 11.9% loss	The 68% end-of-life margin absorbs a large error here

Assumption	Value	Basis	What it drives	If it is wrong
Battery DoD, line efficiency, cell energy, packaging	0.60, 0.90, 150 W·h/kg, 1.5	Cycle-life driven; class figures [3]	38 kg installed battery	Mass is linear in each
Propulsion performance	1,600 s and 40 mN xenon Hall; 220 s hydrazine	Class figures [3]	Propellant budget and servicing capacity	Capacity is roughly linear in I_{sp}
Relocation drift rate	1° per day	Operationally reasonable repositioning speed	5.69 m/s per relocation	Delta-v scales with the rate
RPO allowance	8 m/s per client	Approach, backout, and one contingency abort	Hydrazine budget, which is what binds servicing capacity	Capacity is inversely proportional to it
Component temperature limits	+50 and -20 °C	Conventional electronics band	The margin assessment in Table 9	Sets the stated margins directly
Uncertainty and growth allowances	11 K, 25%	Standard practice before thermal balance test	Radiator sizing trade in Section 8.2	Together they consume the 13.7 K hot margin
Optical properties	α 0.14 / ε 0.03 MLI; α 0.14 / ε 0.78 OSR	Silverized Teflon and optical solar reflector class figures	Equilibrium temperatures	T scales as $(\varepsilon A)^{-1/4}$, so errors are damped
Bus thermal lump	1,400 kg at 900 J/(kg·K), isothermal	Aluminium-dominated dry mass	The 3.9 K eclipse excursion in Section 12.3	Optimistic: real vehicles have gradients, which the Section 8.4 test retires
Total ionizing dose rate	5 krad(Si)/yr behind 100 mil Al	Representative GEO figure	25 krad(Si) mission dose, part class	Category R still holds at twice the assumed rate
Meteoroid density and speed	2.5 g/cm ³ , 20 km/s	Course and textbook conventions [14], [19]	Flux, impact probability, Whipple sizing	Standard values; the flux model is the larger uncertainty
Servicing manifest	242-day cycle, six cycles in five years	Notional, built from the drift, inspection, and mated durations in Figure 1	Mated fraction of 80%, and the split of the delta-v budget	Shifts propellant between the free-flying and mated cases

Three of these deserve more attention than the rest in any follow-on work. The c_{ps} to c_m offset is the single most sensitive number in the report, because the disturbance torque and everything downstream of it are linear in that one estimate. The slew requirement is the basis of the central design claim and is self-levied rather than given. And the servicing manifest sets the propellant split, which is what determines the capacity figure the customer is actually buying.

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